

Article

A Comparison of Machine Learning-Based Approaches in Estimating Surface PM_{2.5} Concentrations Focusing on Artificial Neural Networks and High Pollution Events

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Abstract: Surface PM_{2.5} concentrations have significant implications for human health, necessitating accurate estimations. This study compares various machine learning models, including linear models, tree-based algorithms, and artificial neural networks (ANNs) for estimating PM_{2.5} concentrations using the MERRA-2 dataset from 2012 to 2023. Mutual information and Spearman cross-feature correlation scores are used during feature selections. The performance of models is evaluated using metrics including normalized Nash–Sutcliffe efficiency (NNSE), root mean standard deviation ratio (RSR), and mean percentage error (MPE). Our results show that ANNs outperform linear and tree models, particularly in estimating daily PM_{2.5} concentrations of 35–1000 µg/m³. ANNs improve NNSE by 119% and 46%, RSR by 40% and 24%, and MPE by 44% and 30% from linear and tree models, respectively, indicating ANN’s superior estimation performance during high pollution days. The sensitivity analysis of features that interpret the models suggests that the total extinction AOD at 550 nm and surface CO concentrations are the most important features in the Western and Eastern U.S., respectively. The findings suggest that even the simplest NNs provide better air quality estimates, especially during high pollution events, which is beneficial for long-term exposure analysis. Future research should explore more sophisticated NN architectures with spatial and temporal variations in PM_{2.5} to improve the model performance.

Keywords: machine learning; air quality; artificial neural network; MERRA-2 reanalysis; high pollution events



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1. Introduction

PM_{2.5}, or particulate matter with a diameter less than 2.5 µm, has long been linked to human health and excessive mortality [1]. Epidemiological studies have been conducted to investigate adverse health effects of PM_{2.5}, including respiratory symptoms, cardiovascular complications, and suboptimal health in pregnant women and infants [2,3]. To comprehensively evaluate PM_{2.5} health effects, long-term exposure studies over a large region are needed. Ground monitoring sites can provide high-quality measurements of PM_{2.5} concentration; however, they are inherently limited in terms of geographic coverage. Studies have shown data based on “point” locations cannot accurately represent individual exposure [4].

The two common approaches to overcome the limitations of spatial coverage are chemical transport models (CTMs; e.g., [5]) and statistical models, such as multivariate

linear regression (e.g., [6]). Chemical transport modeling can simulate PM_{2.5} based on emissions and meteorology data incorporating known chemistry mechanisms, and the results can also be constrained by PM_{2.5} measurements from ground sites and remote sensing via data assimilation [7,8]. A downside of conducting CTM simulations, even at a regional scale, is the heavy computational cost depending on factors such as the complexity of the chemistry scheme, the number of tracers included, and the spatial and temporal resolution of the modeling domain [9].

To mitigate the high cost of CTMs, statistical models are often used to estimate surface PM_{2.5} concentration from other measurables. The use of satellite remote sensing measurements, such as the top-of-atmosphere reflectance [10] and aerosol optical depth (AOD; e.g., [11]), are routinely included as covariates to estimate surface PM_{2.5}. Additionally, meteorological and land use variables, which can help characterize the dispersion process of PM_{2.5}, are included to improve model performance [12]. The statistical models have been found to have widely varying performance, critically dependent on the selected covariates. The decision to include or exclude particular predictors (i.e., features/covariates) can often be somewhat arbitrary and not well documented in the literature. While some studies have used as few as three features [13], others have used as many as twenty-seven [14], which demonstrates the lack of a consensus on feature inclusion. Indeed, the importance of these covariates can differ greatly by region and time. Zhan et al. [15] predicted PM_{2.5} in 2014 over China and found that the top three most important features were day of the year, air temperature, and AOD. However, another study [16] using four different machine learning methods found that AOD, visibility, and wind speed were the most important features for Iran from 2011 to 2016. Comparisons across previous studies to obtain a clear recommendation of important features are not always straightforward due to vastly different sets of candidate variables and diverse ranking criteria. Therefore, an objective feature selection approach based on a robust statistics perspective is needed.

While conventional statistical models are widely adopted, machine learning (ML) has recently emerged as a powerful approach for estimating surface PM_{2.5}, given its promising success in many other Earth science fields [17–19]. Chu et al. [6] called for more studies of PM_{2.5} prediction with machine learning because only two studies [20,21] reviewed therein employed machine learning at the time. Since then, there has been a wave of studies using machine learning approaches in the last few years, such as random forest models [22–25] and gradient boosting machines [15].

Among various ML methods, neural networks (NNs) are increasingly used. NNs are composed of multiple layers of interconnected nodes that can learn from data and perform nonlinear transformations [26]. NNs can handle high-dimensional and noisy data and achieve high performance in PM_{2.5} estimation. Many variants of neural networks, such as the three-layer artificial neural networks (ANNs; [27]), the deep learning network [28], and the convolutional neural network [14], have demonstrated success in this research area. However, NNs also have some drawbacks, such as the risk of overfitting, the difficulty of interpretation, and the lack of uncertainty quantification [29]. To overcome the interpretability issue of the NNs, studies have used the permutation feature importance to interpret ANN model results, which involves shuffling each feature in the validation set and measuring how much the shuffle decreases the accuracy of the model. The importance of the shuffled feature is determined by the decrease in the model's performance. If shuffling a feature results in a significant decrease in model performance, that feature is considered important [30]. Conversely, if shuffling has little to no effect on the model's performance, the feature is deemed not important.

Given the fast advance in the development of ML techniques in recent years, fueled by the increasing availability of off-the-shelf packages and open-source ecosystems, several

methodological studies have compared various machine learning algorithms to evaluate their performances. Berrocal et al. [31] compared several linear models (e.g., inverse distance weighting, kriging, and statistical downscaling models) alongside a random forest, a support vector regression (SVR), and a neural network model across the contiguous United States and found the models that took into account spatial information outperformed the ML models; nonetheless, the temporal scale of their study was only in 2011. Zuo et al. [32] compared six different models over the Beijing–Tianjin–Hebei region of China using a (single) decision tree, K-nearest neighbors (KNN), support vector machines (SVMs), random forest, and gradient boost, and they found decision trees outperformed SVMs; however, their models were aimed at classifying different risk levels of PM_{2.5} rather than estimating the actual concentration values. Moreover, Xu et al. [33] compared models for estimating monthly averaged PM_{2.5} concentration, including multiple linear regression, a Bayesian regularized neural network, an SVM, least absolute shrinkage and selection operator (LASSO) regression, multivariate adaptive regression splines (MARS), a random forest, extreme gradient boost (XGBoost) over British Columbia and found decision trees performed the best. Ghahremanloo et al. [34] evaluated PM_{2.5} estimation over Texas, with a fairly small number of models, including ordinary linear regression, a linear mixed-effects model, and a random forest, and found the random forest performed the best. However, a caveat is that the features used by Ghahremanloo et al. [34] included the chemical components that make up the total PM_{2.5} in MERRA-2, such as sea salt, SO₂, and SO₄ concentrations, among others, which contradicts the purpose of estimating PM_{2.5} because knowing these values, one could directly compute the total PM_{2.5}.

There are many ML models available, given their accessibility through open-source codebases. While some favorites have emerged (e.g., the basic random forests and variants of neural networks), it can often be difficult for users to synthesize the results directly across previous studies because the datasets, time, and geographic domains vary greatly [35]. For example, Yan et al. [36] constructed a deep learning model in China but then compared their results to those of other previous studies using different datasets over different years, which is not a straightforward evaluation. Motivated by the need for a more thorough intercomparison, this study aims to provide a comprehensive evaluation of a suite of popular statistical models (including simple and more advanced ML algorithms) to understand and demonstrate their relative strengths and limitations so that future users can make informed decisions.

A systematic comparison of ML algorithms is only meaningful over a consistent spatiotemporal domain and training data. For any model, training data can come from some combination of chemical transport models [37,38], ground-station monitors [24], and satellite data [39]. The latter two, despite being direct observations, are inherently limited by data availability (e.g., because of missing or suspicious values of AOD when retrieved over bright surfaces or during cloudy days [40–42]). Many of these data gaps and quality issues were explored by Shin et al. [43], who made an important critique of the model utility: when calibrated models were applied to unseen data, their performance was drastically worse than when applied to in-sample data. Shin et al. [43] also called for more models to adopt a mix of data sources, such as chemical transport models and satellite observations. Here, we carry out this methodological analysis of model intercomparison using NASA's Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) reanalysis dataset alone, purposefully to bypass the data availability and quality issues. MERRA-2 is a synthetic dataset of meteorological and chemical reanalysis, which has been well validated and widely used in many air quality studies [44]. Its representation of surface pollution is on par with CAMS (the latest European chemical reanalysis) and performs better than CAMS in terms of its AOD product [45].

The goal of this study is to demonstrate the relative advantages of the ANN models in estimating daily surface PM_{2.5} concentrations in the contiguous United States, especially during high pollution days. The performance of different machine learning models, including linear models (linear regression, polynomial regression, elastic net, and Bayesian ridge regression), tree models (AdaBoost, random forest, extra trees, gradient boosting, and XGBoost), and ANN models with different hidden layers (ANN1ANN4), is comprehensively compared using the MERRA-2 dataset from 2012 to 2023. An objective feature selection approach combining mutual information and Spearman correlation scores is also implemented to identify the geographic variance of feature importance. In addition, the study explores a sensitivity analysis of features for tree models and ANN models using feature importance and the permutation method, respectively, from a global perspective and explores the local interpretations of the most important features using Shapley Additive Explanations (SHAP). The results demonstrate the encouraging superiority of ANNs in capturing the complex and nonlinear relationships between PM_{2.5} and its influencing factors across seasons and locations. They provide a pathway toward more reliable and accurate air quality estimates for long-term exposure analysis in public health domains.

2. Data and Methods

2.1. Original Dataset

MERRA-2 is an assimilated reanalysis product from the Goddard Earth Observing System (GEOS) [44]. Numerous satellite measurements are incorporated together with a community radiative transfer model (CRTM) to produce atmospheric radiative properties. There is an extensive aerosol product incorporating measurements from MODIS, MISR, AVHRR, and ground measurements from AERONET.

For this study, we used four data products from MERRA-2, version 5.12.4 (M2I1NXASM, M2SDNXSLV, M2T1NXRAD, and M2T1NXAER, with their full names in Table 1) with a spatial resolution of $0.5^\circ \times 0.625^\circ$ [46–49]. One of the four datasets (M2SDNXSLV) is available with a daily resolution, and the other three data products with hourly or 3-hour resolution were averaged to daily values before any analysis.

Table 1. Short names, long names, and brief descriptions of the data products used in this study, obtained from the Global Modeling and Assimilation Office [46–49].

Short Name	Long Name	Brief Description
M2I1NXASM	MERRA-2 inst1_2d_asm_Nx: 2d,1-Hourly, Instantaneous, <i>Single-Level Diagnostics</i>	An instantaneous 2-dimensional hourly data collection, including assimilations of meteorological diagnostic parameters at the single levels
M2SDNXSLV	MERRA-2 statD_2d_slv_Nx: 2d, Daily, <i>Single-Level Diagnostics</i>	A 2-dimensional daily data collection, including daily statistics of temperature and precipitation
M2T1NXRAD	MERRA-2 tavg1_2d_rad_Nx: 2d,1-Hourly, Time-Averaged, <i>Radiation Diagnostics</i>	An hourly time-averaged 2-dimensional data collection, including radiation diagnostics
M2T1NXAER	MERRA-2 tavg1_2d_aer_Nx: 2d,1-Hourly, Time-averaged, <i>Aerosol Diagnostics</i>	An hourly time-averaged 2-dimensional data collection, including assimilated aerosol diagnostics

The total mass concentration of PM_{2.5}, not directly available from MERRA-2, was calculated following Collow et al. [50]:

$$PM_{2.5} = DUSMASS25 + SSSMASS25 + BCSMASS + OCSMASS + 1.375 \times SO4SMASS \quad (1)$$

where $DUSMASS25$ is Dust Surface Mass Concentration— $PM_{2.5}$, $SSSMASS25$ is Sea Salt Surface Mass Concentration— $PM_{2.5}$, $BCSMASS$ is Black Carbon Surface Mass Concentration, $OCSMASS$ is Organic Carbon Surface Mass Concentration, and $SO4SMASS$ is Sulfate Surface Mass Concentration, all from the aerosol product of M2T1NXAER [48].

The $PM_{2.5}$ provided by MERRA-2 has been extensively evaluated and is found to be comparable with ground measurements [51]. Our previous study [52] found that MERRA-2 $PM_{2.5}$ underestimated the surface concentration over India during the early 21st century, potentially due to the lack of direct inclusion of $PM_{2.5}$ surface measurements collected at point locations. The final dataset we used here included all days in the 12 years from 1 January 2012, to 31 December 2023, with 4150 grid cells in total, spanning the geographic region from 125.96° W to 73.929° W and 24.542° N to 49.855° N and a pool of 115 variables related to trace gases, aerosols, meteorology, radiation, as well as a few land features such as surface albedo, surface emissivity, and the zero-plane displacement height (as a measure of topography). For variables of $PM_{2.5}$, total extinction AOD, total absorption AOD, surface CO concentrations, and surface O_3 concentrations, we also performed a natural log transformation because they were right/left skewed. ML models can learn efficiently when the data follow a normal distribution, which is a data scaling step we often use during the model set-up.

2.2. Methods of Selecting Features from the Original Dataset

2.2.1. Mutual Information

Mutual information (MI) is a measure of the amount of information shared between two variables. MI quantifies the reduction in uncertainty about one variable, given the knowledge of the other variable. MI is defined as

$$I(X; Y) = \int_y \int_x P_{(X,Y)}(x, y) \log \left(\frac{P_{(X,Y)}(x, y)}{P_X(x)P_Y(y)} \right) dx dy \quad (2)$$

where X and Y are two discrete random variables, $P_{(X,Y)}(x, y)$ is the joint probability distribution of X and Y , and $P_X(x)$ and $P_Y(y)$ are the marginal probability distributions of X and Y , respectively.

MI can be used to select the most relevant features for estimating $PM_{2.5}$ concentration. MI can capture both linear and nonlinear dependencies between variables, unlike Pearson correlation coefficients, which only measure linear relationships. By ranking the features according to their MI with the target variable ($PM_{2.5}$ concentration), we can select the most informative features and reduce the dimensionality of the data. For example, Pak et al. [53] used MI to perform the spatiotemporal correlation between $PM_{2.5}$ and observation variables, considering their linear and nonlinear relationships.

2.2.2. Spearman Correlation

The Spearman correlation is a nonparametric measure of rank correlation that evaluates the degree of association or correlation between two variables using the ranks of their data rather than the raw data themselves. It is particularly useful for understanding relationships between variables when the data do not meet the assumptions necessary for Pearson's correlation coefficient, such as linearity and normal distribution [54]. The term nonparametric means that the Spearman correlation does not assume a specific distribution for the data, making it robust against outliers and suitable for ordinal data or nonlinear relationships.

Spearman's correlation coefficient is defined as the correlation coefficient between the ranked variables, which refers to variables that have been arranged or sorted according to their value or magnitude, assigning each rank based on its position in the sorted order. It is calculated by ranking each set of data and then determining the correlation coefficient based on the differences between these ranks [55]. For a sample of size n , the n raw observations are converted to ranks, and r_s is computed as

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (3)$$

where d is the difference between the ranks of each observation, and $\sum_{i=1}^n d_i^2$ is the sum of the squared differences between ranks. While $r_s = 0$ means there is no monotonic relationship between variables, $r_s = 1$ (or -1) represents a perfect positive (or negative) monotonic relationship. However, it is important to note that high correlation does not necessarily imply a direct influence on the model's performance.

The Spearman correlation is appropriate for data that follow curvilinear, monotonic relationships. The widely used Pearson correlation, on the other hand, is only valid for data that follow linear relationships [55]. Pearson correlation may not capture the full relationship between meteorological factors, trace gas features, aerosol optical depths, and spatial/temporal features, while the Spearman correlation is more robust when data are non-normally distributed [56].

2.3. Methods of Splitting Dataset After Feature Selections

After preprocessing the MERRA-2 dataset and selecting key variables related to PM_{2.5} for model inputs, we divided the data into three sets with a 4:1:1 ratio: a training set with eight years of data (2012–2019), a validation set with two years of data (2020–2021), and a test set with two years of data (2022–2023). While randomly splitting the dataset at a certain ratio can enhance model performance, we chose to split the data in chronological order to prevent the models from having any knowledge of future years in the current changing climate. The validation set serves a critical purpose: it allows for the efficient tuning of hyperparameters during model training, and it validates the trained models to mitigate the risks of overfitting and underfitting. One might question the decision against implementing the 10-fold cross-validation method; this is primarily due to the substantial size of our dataset ($\sim 365 \times 4150 \times 12$ data points), making cross-validation excessively time-consuming. We trained linear, tree, and ANN models using the training set and calculated the performance metrics introduced in Section 2.5 with the validation set. This training process was repeated until performance improvements became marginal (less than a 1% difference in the performance metrics compared to previous results). Following model training, we estimated PM_{2.5} concentrations on the test data and documented the final performance metrics.

2.4. Methods of Machine Learning Models Setup and Hyperparameters

Table 2 shows a brief overview of the machine learning models and the research on their applications in the area of air quality. In addition to the models used in studies reviewed in the introduction section, we also evaluated models such as polynomial regression, elastic net, Bayesian ridge regression, and Adaptive Boosting.

The specific model setup, relevant studies reviewed in Table 2, and hyperparameters used when training the models are detailed in Sections 2.4.1–2.4.10.

Table 2. A summary of the machine learning models evaluated in this study, along with 1–4 relevant studies that apply the model in air quality predictions.

Model Names	Brief Overview	Relevant Studies
Linear regression	A statistical method that models the relationship between a dependent variable and one or more independent variables as a straight line.	[31,33,34,40]
Polynomial regression	An extension of linear regression where the relationship between variables is modeled as an n th-degree polynomial.	[57,58]
Elastic net	A regularized regression model combining L1 (Lasso) and L2 (ridge) penalties to handle multicollinearity and variable selection.	[59]
Bayesian ridge	A regression model using Bayesian principles to estimate the posterior distribution of model coefficients, incorporating uncertainty.	[60]
Adaptive Boosting (AdaBoost)	An ensemble learning technique that combines weak learners sequentially, giving more weight to misclassified samples in each iteration.	[61,62]
Random forest	A tree-based ensemble model that builds multiple decision trees and aggregates their predictions for robust performance.	[23–25,31]
Extra trees	Similar to random forest but splits nodes randomly instead of optimally, improving computation speed and reducing overfitting.	[63]
Gradient boost	An ensemble technique that builds models iteratively, optimizing for a loss function by fitting new models to the residual errors.	[15,16,23,32]
Extreme Gradient Boost (XGBoost)	An efficient implementation of gradient boosting optimized for speed and performance, with features like regularization and parallelism.	[33,61]
Artificial neural network with 3 hidden layers (ANN1)	The simplest form of deep learning models that processes data through multiple nonlinear transformations across hidden layers to learn complex patterns.	[27,31,61]
Artificial neural network with 4 hidden layers (ANN2)		
Artificial neural network with 5 hidden layers (ANN3)		
Artificial neural network with 6 hidden layers (ANN4)		

2.4.1. Linear Regression

Linear regression minimizes the sum of the squared residuals (ordinary least squared) to identify the line of best-fit and coefficients for each feature. This was included here as a baseline for performance and computational efficiency, and it was expected any model that can account for a nonlinear relationship would outperform it. The version of linear regression used is the scikit-learn version of 1.5.2.

2.4.2. Polynomial Regression

Handling the intrinsic nonlinear relationship between the independent variables (such as AOD, meteorological features, and land features) with $PM_{2.5}$ can be achieved with a model specifically taking into account these interactive relationships by creating a polynomial combination of the features. For instance, Bui et al. [57] used polynomial regression as a kernel function in their models for predicting PM_{10} concentration from

drilling operators. James et al. [58] used polynomial functions to fit in each grid cell to provide the nonlinear response of monthly average PM_{2.5} to changes in NO_x, SO₂, VOC, NH₃, and primary PM_{2.5} emissions across their domain. To avoid overfitting, here, only the second-order polynomials were included in addition to the linear term, including the interactive terms, such as the product of two distinct features. The version of polynomial regression used is the scikit-learn version of 1.5.2.

2.4.3. Elastic Net

In terms of the elastic net regression model, Gao et al. [59] used it to estimate the asthma prevalence rate at different ages using air pollution, weather data, and asthma data from 2015 to 2021, which encouraged us to apply it in exploring the relationship between PM_{2.5} and other influential factors. To avoid overfitting and improve out-of-sample performance, it is standard practice to apply a penalty to the coefficients in a linear model, thereby enhancing its ability to generalize. This technique, known as regularization, involves minimizing the sum of the squared residuals along with an additional penalty term. There are two main types of regularization, detailed below. The double bars represent the p-norm of a vector.

$$\|\vec{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \quad (4)$$

$$\text{L1 regularization, which minimizes } \|Y - X\omega\|_2^2 + \alpha \|\omega\|_1 \quad (5)$$

$$\text{L2 regularization, which minimizes } \|Y - X\omega\|_2^2 + \alpha \|\omega\|_2^2 \quad (6)$$

where ω denotes the coefficients of the linear model, and λ is the regularization parameter, which was determined through cross-validation. Both regularization methods penalize the coefficients—the elements of w —to find values that optimize out-of-sample performance. L1 regularization (Least Absolute Shrinkage and Selection Operator, LASSO regression) typically results in models with fewer non-zero coefficients, while L2 regularization (ridge regression) generally produces models with smaller coefficients.

Combining LASSO and ridge regression yields a model that balances both sparsity and coefficient size, known as elastic net [64]. An elastic net employs both L1 and L2 regularization as follows:

$$\frac{1}{2n_{\text{sample8}}} \|Y - X\omega\|_2^2 + L1_{\text{ratio}} \alpha \|\omega\|_1 + \frac{\alpha}{2} (1 - L1_{\text{ratio}}) \|\omega\|_2^2 \quad (7)$$

In this context, the $L1_{\text{ratio}}$ determines the relative influence of L1 versus L2 regularization. This model was selected to explore whether regularizing a basic linear model while keeping all other factors constant would enhance its generalization capabilities. The elastic net (with the scikit-learn version of 1.5.2) was tuned across the entire training dataset to determine the optimal $L1_{\text{ratio}}$ of 0.1—the parameter that decides the extent of L1 and L2 regularization—and the α value of 0.1.

2.4.4. Bayesian Ridge

Bayesian ridge regression estimates the maximum likelihood function of the coefficients of a linear model (thus Bayesian) and also includes an L2 regularization estimated by maximizing the log marginal likelihood [65]. A Gaussian prior is used for the parameters of the likelihood function, and the posterior parameters can be estimated [66]. The regularization parameter is derived directly from the data, bypassing the need for cross-validation. In terms of its application in the air quality area, Plocoste and Laventure [60] used Bayesian ridge regression to predict PM₁₀ concentrations in the Caribbean area with PM₁₀, tempera-

ture, and rainfall as the input parameters. The version of Bayesian ridge regression used is the scikit-learn version of 1.5.2.

2.4.5. Adaptive Boosting (AdaBoost)

AdaBoost is an algorithm that learns to combine many weak estimators to produce a more powerful estimator [67]. The base weak estimator used here was chosen through hyperparameter tuning from estimations on the validation set. The idea behind AdaBoost is to improve each successive estimator based on the error produced by the last estimator trained. This learning is refined iteratively until sufficiently small errors are produced with the newly added stump or the maximum number of trees is reached.

In addition, Zhai and Chen [61] used Adaptive Boosting, XGBoost, and ANN models to forecast and analyze the daily average PM_{2.5} concentrations in Beijing from 2013 to 2017. The authors asserted that the ANN exhibited superior performance compared to AdaBoost and XGBoost in estimating the daily PM_{2.5} concentrations. Chen et al. [62] also employed the Adaptive Boosting model and XGBoost to estimate the hourly PM_{2.5} concentrations in China in 2016, but they did not explore the application of ANN models.

The final hyperparameters chosen by the validation set include a base estimator with five trees, a learning rate of 0.3, an exponential loss, and 200 estimators. The version of AdaBoost used is the scikit-learn version of 1.5.2.

2.4.6. Random Forest

A random forest is an ensemble of trees, each built with a random subset of features from a random subset of the training dataset [68]. The number of trees used to make the forest is an arbitrary hyperparameter but can be tuned through the validation set. In this study, we used 80 estimators, a min sample split of 2, a min number of samples per leaf of 2, a max number of features of $\log_2$, and a `ccp_alpha` of 0 to train the models (with the scikit-learn version of 1.5.2).

Many stochastic decisions are used in the creation of a random forest, hence the name. To start, a random subset of the training data is used as a bootstrapped batch. Then, to build a tree, a random subset of the features of the bootstrapped dataset is picked to determine which feature would be the “best” to reduce the uncertainty in the estimation using the Gini impurity index. The depth of each tree then varies based on the subset of data used for that tree, the parameters limiting the depth of the tree, and the number of samples required for the algorithm to split the data.

2.4.7. Extra Trees

Extra trees are trained in much the same way as a random forest. However, the threshold used to split at the nodes of the tree is also randomly chosen, meaning the entire process of building the trees is random. Geurts et al. [69] stated that this extreme randomization can greatly reduce the out-of-sample variance. Yu et al. [63] also used extra trees to estimate the PM_{2.5} concentration over the continental U.S. using GOES-16 AOD, but the authors claimed that there was a large potential to improve their results, such as more input variables and the model performance. In this study, we used 70 estimators, a min number of samples per leaf of 2, and a `ccp_alpha` of 0 to train the models (with the scikit-learn version of 1.5.2).

2.4.8. Gradient Boost

Gradient boost is an approximation technique developed by Friedman [70]. Gradient boost begins by making an initial guess of what the target value should be, usually the simple average. Then, a residual is computed between each of the estimated values and the earlier guess. From then on, a new tree is built to estimate these residuals, thereby

improving the guess for each input vector. After each tree is built, the residuals of the estimation are computed again, and the next tree is built to estimate these new residuals. This process continues until the number of trees allowed is exhausted or there is a near-perfect fit. Gradient boost (with the scikit-learn version of 1.5.2) contains three regularization parameters to prevent overfitting: the number of trees, the depth of the trees, and the learning rate (held the same for each tree in this study). The learning rate scales the output of each tree such that there is always a residual, meaning that a gradient always exists to allow the next tree to refine the guess to prevent overfitting.

2.4.9. Extreme Gradient Boost (XGBoost)

XGBoost is similar to Gradient Boost, but it includes several architectural and algorithmic changes, which allow the algorithm to scale well to run on parallel and distributed systems [71]. Among these changes is the algorithm that decides what split to use for a given feature at any given node that refers to a decision point within a decision tree as part of the ensemble. At each node, the model considers a split based on a specific feature to divide the data into two groups and to improve the purity of the target variable within those groups. The algorithm's job is to determine the best feature to split on at each node, considering the potential reduction in a given loss function. XGBoost also randomizes the training data for each tree, which can reduce overfitting. The version of the XGBoost used is 2.1.1. The hyperparameters of the model included a subsample of 0.9, a `colsample_bytree` of 0.5, a `max_depth` of 5, a learning rate of 0.1, a `gamma` of 0, and a `reg_lambda` of 0.

2.4.10. Artificial Neural Networks

Neural networks have rapidly taken prominence as one of the most successful forms of machine learning over the past decade. The building block of a neural network is the perceptron (also known as neuron/node/unit), a mathematical unit that computes a dot product between an input vector and a set of weights/coefficients. The basic form of neural networks, like artificial neural networks, consists of one or more layers of perceptrons connected by a matrix of weights and biases to the input features, which produces a number at each node.

The models learn to fit a nonlinear function by taking small steps along a gradient, measuring the error of the current model parameters. This is a process known as backpropagation and is used to decide the optimal weights comprising the final neural network. A critical component of backpropagation is the activation function, which enables the optimization of a neural network. The architecture of a neural network can be flexible, ranging from simple, one-layer networks to networks hundreds of layers deep. Further, the connections within and between the layers can take on complicated forms, allowing networks to remember past values (e.g., long short-term memory networks, attention mechanisms) or even have connections skip forward one or more layers (i.e., skipped connections).

Because of the vast complexity available, here, we only showcased a few configurations of ANNs (with the TensorFlow version of 2.17.0) to evaluate the sensitivity of performance after the hyperparameter tuning. The architectures we tested are listed below:

- ANN1: 3 hidden layers with 3536 neurons in each layer
- ANN2: 4 hidden layers with 3536 neurons in each layer
- ANN3: 5 hidden layers with 3536 neurons in each layer
- ANN4: 6 hidden layers with 3536 neurons in each layer

Note at the end of the last hidden layer, all of the nodes were combined into the single final estimations of $PM_{2.5}$ for each location and each day.

2.5. Methods of Evaluating Model Performance

The correlation coefficient (R^2), the normalized Nash–Sutcliffe model efficiency coefficient (NNSE), the root-mean-square error (RMSE), the root mean standard deviation ratio (RSR), the mean absolute error (MAE), the mean percentage error (MPE) and the symmetric mean absolute percentage error (SMAPE) are metrics used to compare various algorithms under different scenarios, especially for the selection of MPE or SMAPE to compare model performance. For example, when comparing the national model performance on the linear scale, it is better to choose SMAPE because MPE can give a misleading impression of model performance if the errors are balanced but large when positive and negative errors can cancel each other out. Equations (8)–(14) show the calculations of these metrics, respectively.

$$R^2 = 1 - \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (8)$$

$$NNSE = \frac{1}{1 + \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2}} \quad (9)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - O_i)^2} \quad (10)$$

$$RSR = \frac{RMSE}{\sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - \bar{O})^2}} \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_i - O_i| \quad (12)$$

$$MPE = \frac{1}{n} \sum_{i=1}^n \left(\frac{P_i - O_i}{O_i} \right) \times 100 \quad (13)$$

$$SMAPE = \frac{1}{n} \sum_{i=1}^n \left(\frac{|P_i - O_i|}{|P_i| + |O_i|} \right) \times 100 \quad (14)$$

where O_i are the observed values, P_i are the estimated values, $\bar{O}$ is the mean of the observed values, and n is the number of observations.

Specifically, R^2 measures how well the regression model fits the observed data. It ranges from 0 to 1, with higher values indicating a better fit. The R^2 value can be interpreted as the proportion of the variance in the dependent variable that the independent variables in the model can explain.

The NNSE further measures how well the plot of observed versus estimated data fits the 1:1 line (unity line). A value of 1 indicates a perfect fit, and 0 means the model estimations are as accurate as the mean of the observed data.

In addition, the MAE and RMSE quantify the size of the residuals of the estimations on the dataset, and lower values indicate better performance. The values of MAE and RMSE reported here are in units of $\mu\text{g}/\text{m}^3$. The MAE is the average of the absolute values of the errors or residuals. It represents the average magnitude of the estimation errors, regardless of their direction. The RMSE is the square root of the average of the squared errors or residuals. It represents the standard deviation of the estimation errors, which measures how much the estimations deviate from the actual values.

Moreover, the MPE, SMAPE, and RSR are used to quantify the normalized error of machine learning models. The MPE indicates the average percentage difference between estimated and observed values. A value close to 0 is ideal, with negative values indicating underestimation and positive values indicating overestimation on average. The SMAPE provides a symmetric, more balanced version of the MPE by using the absolute value of the difference and the average of the absolute actual and predicted values. This symmetrically

accounts for both over- and underestimations and avoids canceling out errors, as discussed before. The range of SMAPE is between 0 and 100%, where 0% indicates a perfect estimation. In this study, the SMAPE was used when comparing the individual and average model performance on most days, while the MPE was used when comparing the average performance on high pollution days when the daily $\text{PM}_{2.5}$ concentration was larger than $35 \mu\text{g}/\text{m}^3$ and regional performance at each grid cell. The RSR normalizes the RMSE by the standard deviation of observed data, whose values closer to 0 indicate a better model, with 0 being the optimal value.

3. Results

3.1. Feature Selections as Model Inputs

Out of the 112 available variables initially obtained from the MERRA-2 dataset (Table 1), estimates of the column burden of aerosols were dropped first because we wanted to include only features that could be robustly and directly measured and constrained. We also excluded features such as the daily minimum and maximum temperature, as well as radiative features like the top-of-atmosphere outgoing longwave flux, which had clear duplicates with other features. After the initial screening, we identified a long list of 24 features, consisting of 18 meteorological features, 2 trace gas features, and 4 spatial/temporal features. We then applied Spearman's cross-feature correlation and the mutual information (MI) scores between each candidate feature and the target variable, $\text{PM}_{2.5}$, to subsequently select highly relevant features without covariance. The detailed procedure was as follows:

1. Compute the MI between all 24 features and the target variable $\text{PM}_{2.5}$, then rank the features by their MI scores. Compute the Spearman cross-feature correlation scores between the dependent variables. Note that $\text{PM}_{2.5}$ is included in the correlation plot for reference, but the main purpose of the plot is to filter features with covariance.
2. Start from the feature with the highest MI scores from the ranking list in step 1:
 - a. If the absolute value of the Spearman correlation with any feature already in the final selected features list is equal to or greater than 0.85, skip this feature to avoid strong covariance. Otherwise, add it to the list of final selected features.
 - b. Continue this process until the total number of final selected features reaches 15 or there are no more features left in the ranking list.

The MI scores and Spearman cross-feature correlation results of the final 15 selected features are shown in Table 3 and Figure 1.

Table 3. Mutual information scores (MI Score), mean values (Mean), and standard deviation (Std) of 15 final selected features, along with their long names and units, after filtering the features with strong covariances. The MI Score measures the relevance between features and the target variable ($\text{PM}_{2.5}$) by quantifying how much information one feature provides about $\text{PM}_{2.5}$. The 15 final selected features include 2 trace gases features, 10 meteorological features, plus 2 spatial features and 1 temporal feature.

Feature Names	Long Name	MI Score	Mean	Std	Units
TOTEXTTAU	Total aerosol extinction AOD at 550 nm	0.36	0.13	0.11	ln(unitless)
TOTABSTAU	Total aerosol absorption AOD at 550 nm	0.26	0.01	0.01	ln(unitless)
QV2M	2 m specific humidity	0.18	8.05	4.92	g kg^{-1}
T2M	2 m air temperature	0.17	286.96	11.15	K
COSC	Surface CO concentration	0.16	0.13	0.10	ppmv
PS	Surface pressure	0.11	949.59	73.75	hPa

Table 3. Cont.

Feature Names	Long Name	MI Score	Mean	Std	Units
MonthCos	Cosine component of the month	0.08	0.00	0.71	unitless
ALBEDO	Surface albedo	0.08	0.17	0.09	unitless
lat	Latitude	0.07	37.25	7.22	degrees_north
EMIS	Surface emissivity	0.06	0.98	0.01	unitless
DISPH	Zero-plane displacement height	0.06	2.28	3.67	m
lon	Longitude	0.06	−100.00	14.97	degrees_east
V2M	2 m northward wind	0.04	−0.20	2.47	m s ^{−1}
SLP	Sea-level pressure	0.03	1015.31	6.48	hPa
O3	Surface O ₃ concentration	0.02	40.88	6.93	ppbv

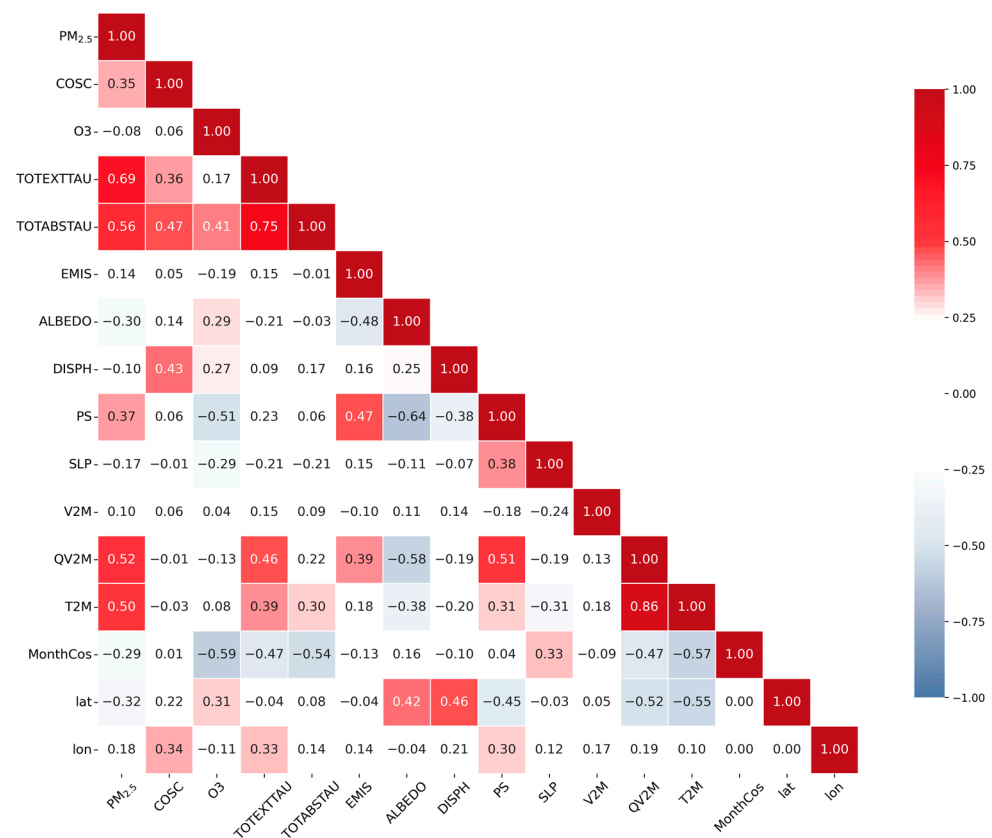


Figure 1. Spearman cross-feature correlation scores of 15 final features after filtering the features with strong covariances, except for the first column that indicates the correlation between features and the target variable ($PM_{2.5}$). Note that $PM_{2.5}$ is included in the correlation plot for reference, but the main purpose of the plot is to filter features with covariance. The correlation score of 0 means there is no monotonic relationship between features. Features with correlation scores closer to 1 or −1 are considered to have a stronger monotonic relationship among them.

The final list of 15 selected features in Table 1 includes MERRA-2's short variable names and full descriptive names. We also computed the Spearman cross-feature correlation with the final selected features after filtering the features with strong covariances in Figure 1. Although some features were still highly correlated (e.g., temperature and specific humidity), they could not be ignored due to our state knowledge of atmospheric science. Note that the candidate features also included the spherical spatial coordinates

(latitude and longitude). In addition, the temporal components we considered were the value of the month, and we transformed it circularly, placing the values 1 (for January) and 12 (for December) close to each other numerically.

$$MonthCos = \cos(2 \times \pi \times Month\ Number / 12) \quad (15)$$

$$MonthSin = \sin(2 \times \pi \times Month\ Number / 12) \quad (16)$$

The top features defined solely by MI included total extinction and absorption AOD at 550 nm, 10 m specific humidity, 10 m air temperature, and surface CO concentration. The AOD is a measure of the light extinction caused by aerosols, which stands out clearly as the most important one, consistent with many previous studies. Higher specific humidity can affect the secondary formation of aerosols in the lower atmosphere, which increases the surface PM_{2.5} concentration. Moreover, the surface CO concentrations also help produce secondary aerosols (e.g., by their reactions with hydroxyl (OH) radicals in the atmosphere). The list also includes the surface ozone concentrations, the cosine component of the month in a year, as well as land properties of surface albedo (potentially as a measure of land type) and zero displacement height (as a measure of surface topography and roughness). Following the simple two-step approach above utilizing MI and Spearman cross-feature correlations, all the trace gas, temporal, spatial, and meteorological features were chosen from a relatively objective data-centric method rather than hand-picked based on physical intuition [31,72,73]. For example, Hu et al. [22] used a total of 38 features (even including variables related to the emission and removal of PM_{2.5}) but without accounting for their collinearity. Zhan et al. [15] used only nine features but without an explicit test of their feature importance.

3.2. PM_{2.5} Estimation Based on Various ML Models

Before computing the performance metrics of machine learning models, it is beneficial to plot histograms and kernel density estimates to visualize data distribution and evaluate the accuracy of the estimations relative to the actual data. Moreover, if the estimations exhibit long tails or significant skewness, metrics such as the RMSE and RSR might be disproportionately influenced by extreme values. In such cases, using metrics like the MAE, MPE, or SMAPE could be more appropriate.

Figure 2a shows the kernel density distributions of the actual PM_{2.5} values (bottom green curve) compared to those estimated by each model. The linear models (blue curves) tended to have higher overestimations. The tree models (purple curves), except for AdaBoost, generally had smooth shapes and captured the full range of PM_{2.5} concentrations better than the linear models. The artificial neural networks (red curves) had shapes much closer to the true values. However, the discrepancies between individual models, except AdaBoost, and actual values were not obvious on the ln scale.

To comprehensively compare the model performance, Figure 2b provides the histogram density distribution of estimates from averages of linear models, tree models (excluding AdaBoost), and ANN models on a linear scale. All three model categories tended to overestimate PM_{2.5} during low and moderate pollution days (0–35 µg/m³) and underestimated PM_{2.5} during high pollution days (>35 µg/m³), where 35 µg/m³ is the daily threshold concentration currently used to determine compliance with the Clean Air Act standards from U.S. Environmental Protection Agency [74]. Specifically, the overestimation during low and moderate pollution days by ANN models was less than that of tree models and linear models. While the underestimation by each model category is challenging to compare with the actual values in Figure 2b, Figure 2c further depicts the KDE plot with

axes on the log10 scale for $PM_{2.5}$ concentrations ranging from 1 to 100 $\mu g/m^3$. This allows for a comparison of model performance during low and high pollution days. In Figure 2c, it is evident that the ANN curve aligns more closely with the actual $PM_{2.5}$ curve during both low and high pollution days, especially at the heads and tails of the curves.

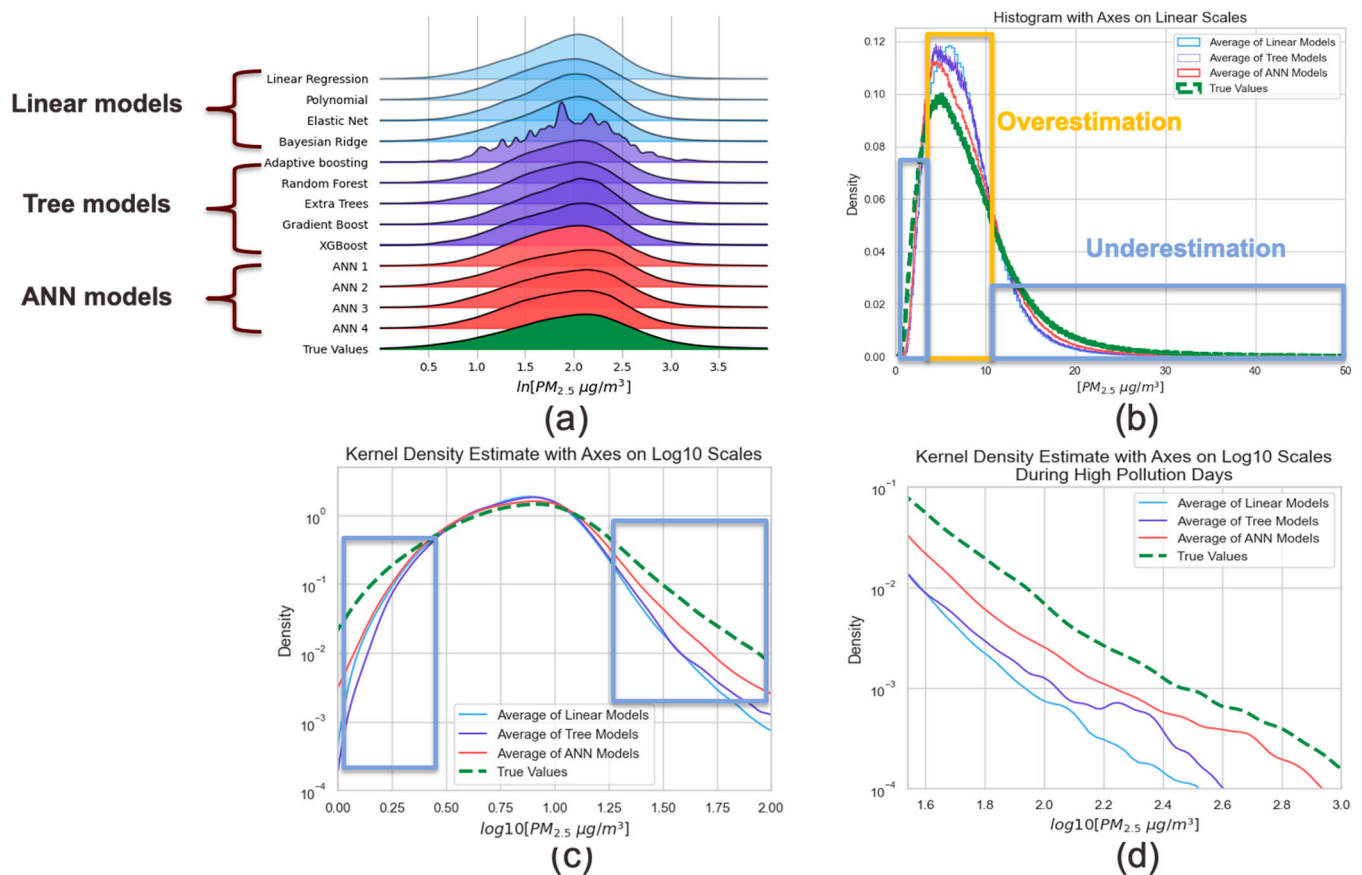


Figure 2. (a) Kernel density distribution of $PM_{2.5}$ concentration estimated on the test set by individual ML models and the actual values in MERRA-2 from 0 to 50 $\mu g/m^3$ on the ln scale. (b) histogram density distribution of estimates from the averages of each model category and the actual values in MERRRA-2 from 0 to 50 $\mu g/m^3$; (c) kernel density distribution of estimates from the averages of each model category and the actual values in MERRRA-2 from 1 to 100 $\mu g/m^3$ on the log10 scale; (d) kernel density distribution of estimates from the averages of each model category and the actual values in MERRRA-2 during high pollution days from 35 to 1000 $\mu g/m^3$ on the log10 scale. In panel (a), the blue, purple, red, and green colors represent linear models, tree models, ANN models, and true values, respectively. In panels (b–d), the blue, purple, and red bins and lines represent the average estimates of linear models, tree models (excluding AdaBoost), and ANN models, respectively; the dashed green bin and line represent true values. In panels (b,c), the yellow box represents the area where models overestimated the actual $PM_{2.5}$ concentrations, while the dark blue box represents the area where models underestimated the actual $PM_{2.5}$ concentrations.

For a further comparison of model performance during high and extremely high pollution days, which is important for air quality and epidemiology studies, Figure 2d shows the KDE plot for $PM_{2.5}$ concentrations from 35 to 1000 $\mu g/m^3$ on the log10 scale. The upper limit concentration of 1000 $\mu g/m^3$, used in this study, is a subjective and arbitrary value primarily intended for visualization purposes rather than as a strict upper bound. Figure 2d clearly indicates that ANN models consistently performed the best, while linear models exhibited the poorest performance in estimating $PM_{2.5}$ during high pollution days.

Overall, Figure 2a–d indicate that the distribution of estimated and actual $PM_{2.5}$ values exhibited a long tail, and using a log10 scale was more effective for comparing

model performance at the distribution's tail. Additionally, the average estimations from ANN models consistently aligned more closely with the true PM_{2.5} concentrations across low, moderate, and high pollution days. Nonetheless, computing performance metrics remained important for comprehensively comparing the various machine learning models.

3.3. Comparison of Individual Model Performance

In assessing the performance of various ML models for PM_{2.5} estimation, Table 4 shows the strengths and weaknesses of each approach through the performance metrics introduced in Section 2.5. To ensure a comprehensive and meaningful comparison of performance metrics, we calculated R² and NNSE on log10-transformed datasets on most days to stabilize variance and mitigate the impact of outliers, leading to more robust and consistent metrics. In contrast, RMSE, RSR, MAE, and SMAPE were calculated on the original datasets on most days to provide intuitive and straightforward interpretations of error magnitudes and percentages in the original data scale. This dual approach balanced statistical robustness and practical interpretability, offering a nuanced assessment of model performance across different data aspects.

Table 4. The performance metrics, including R², RMSE, and MAE, of models over the validation and test datasets. The best-performing models in each model category are shown in bold.

Model Names	R ² _Validation	RMSE_Validation (µg/m ³)	MAE_Validation (µg/m ³)	R ² _Test	RMSE_Test (µg/m ³)	MAE_Test (µg/m ³)
Linear regression	0.65	7.10	3.20	0.65	14.25	2.81
Elastic net	0.71	6.05	2.92	0.71	5.33	2.57
Polynomial	0.63	6.88	3.23	0.64	8.98	2.81
Bayesian ridge	0.65	7.10	3.20	0.65	14.25	2.81
AdaBoost	0.69	6.15	3.05	0.69	5.15	2.65
Random forest	0.78	5.37	2.59	0.78	4.53	2.23
Extra trees	0.78	5.35	2.59	0.78	4.50	2.22
Gradient boost	0.72	5.80	2.86	0.72	5.03	2.51
XGBoost	0.75	5.58	2.75	0.75	4.81	2.39
ANN1	0.80	4.94	2.48	0.79	4.13	2.14
ANN2	0.80	4.89	2.44	0.80	4.05	2.10
ANN3	0.80	4.80	2.42	0.80	4.01	2.09
ANN4	0.80	4.94	2.46	0.80	4.10	2.12

Specifically, Table 4 shows the performance metrics of R², RMSE, and MAE over the validation and test datasets. In the validation set that was used for hyperparameter training and model validation, elastic net regression performed the best among linear models (blue) with an R² of 0.71, RMSE of 6.05 µg/m³, and MAE of 2.92 µg/m³; extra trees provided the best estimation in the tree models (purple) with an R² of 0.78, RMSE of 5.35 µg/m³, and MAE of 2.59 µg/m³; and ANN3 showed the best performance in the ANN models (red) with an R² of 0.80, RMSE of 4.80 µg/m³, and MAE of 2.42 µg/m³. When comparing the top-performing model in each category, the ANN models outperformed both tree-based and linear models.

On the test dataset, which consists of out-of-bag samples, elastic net regression was again the best performer among linear models, with an R² of 0.71, RMSE of 5.33 µg/m³,

and MAE of $2.57 \mu\text{g}/\text{m}^3$. The extra trees algorithm continued to excel among the tree-based models, with an R^2 of 0.78, RMSE of $4.50 \mu\text{g}/\text{m}^3$, and MAE of $2.22 \mu\text{g}/\text{m}^3$. In the ANN category, ANN3 demonstrated the highest performance, with an R^2 of 0.80, RMSE of $4.01 \mu\text{g}/\text{m}^3$, and MAE of $2.09 \mu\text{g}/\text{m}^3$. Overall, ANN models also outperformed the tree-based and linear models on the test dataset.

In addition, a comparison of R^2 values between the validation and test datasets showed that the scores did not vary significantly, indicating there were no overfitting or underfitting issues during model training. The RMSE values in the validation dataset were slightly higher than those in the test dataset because the average concentrations of $\text{PM}_{2.5}$ in the validation dataset were also higher. Therefore, using normalized performance metrics like NNSE and RSR, as shown in Table 5, helped to compare different datasets and mitigated the impact of outliers. Additionally, the RMSE was expected to be higher than the MAE because the datasets exhibited a long-tail distribution, as discussed in Section 3.2.

Table 5. The normalized performance metrics, including NNSE and RSR, as well as the SMAPE, of individual models over the validation and test datasets. The best-performing models in each model category are shown in bold.

Model Name	NNSE_Validation	RSR_Validation	SMAPE_Validation (%)	NNSE_Test	RSR_Test	SMAPE_Test (%)
Linear regression	0.74	0.82	15.20	0.74	1.94	14.50
Elastic net	0.78	0.70	13.92	0.78	0.73	13.41
Polynomial	0.73	0.80	15.28	0.74	1.22	14.55
Bayesian ridge	0.74	0.82	15.20	0.74	1.94	14.50
AdaBoost	0.77	0.71	14.61	0.76	0.70	14.07
Random forest	0.82	0.62	12.34	0.82	0.62	11.72
Extra trees	0.82	0.62	12.33	0.82	0.61	11.67
Gradient boost	0.78	0.67	13.78	0.78	0.69	13.30
XGBoost	0.80	0.65	13.26	0.80	0.65	12.61
ANN1	0.83	0.57	12.12	0.83	0.56	11.53
ANN2	0.83	0.57	11.94	0.83	0.55	11.28
ANN3	0.84	0.56	11.89	0.84	0.55	11.29
ANN4	0.83	0.57	12.09	0.83	0.56	11.43

Furthermore, Table 5 provides the performance metrics NNSE, RSR, and SMAPE for the validation and test datasets. In the validation set, elastic net regression achieved the best performance among linear models, with an NNSE of 0.78, RSR of 0.70, and SMAPE of 13.92%. Among tree-based models, the extra trees model performed best with an NNSE of 0.82, RSR of 0.62, and SMAPE of 12.33%. In the ANN category, ANN3 showed the highest performance, with an NNSE of 0.84, RSR of 0.56, and SMAPE of 11.89%. Comparing the top models in each category revealed that ANN models outperformed both tree-based and linear models.

On the test dataset, elastic net regression again led among linear models, with an NNSE of 0.78, RSR of 0.73, and SMAPE of 13.41%. The extra trees model was the best among tree-based models, with an NNSE of 0.82, RSR of 0.61, and SMAPE of 11.67%. ANN3 outperformed other ANN models, with an NNSE of 0.84, RSR of 0.55, and SMAPE of 11.29%. When comparing the best model in each model category, the ANN3 model

outperformed extra trees and elastic net regression models, with the difference being more pronounced when comparing RSR values (~25% improvement from elastic net and ~10% improvement from extra trees).

Overall, the ANN models exhibited consistency in their performance, reflecting a solid grasp of the data's variance. Their performance metrics scores remained better across different architectures, indicating strong estimation accuracy that was not sensitive to the specific neural network setup. The Methods section detailed the distinctions between neural network versions, and their collective performance highlights the adaptability of basic neural network structures in estimating PM_{2.5} concentrations, exceeding the widely used tree models. Hopefully, using neural network models with spatial information (e.g., convolutional neural network) and temporal information (e.g., long-short term memory) could yield even more precise estimations with distinctly higher R² and lower RMSE values.

3.4. Comparison of Average Model Performance

In this section, we further evaluate the accuracy of various machine learning models for estimating PM_{2.5} concentrations by comparing estimated values against actual values with scatter plots under three scenarios (panels a–c in Figure 3). These scenarios correspond to those in Figure 2b–d. Instead of highlighting the best model in each category, we chose to show the average performance of the linear, tree, and ANN models because that approach provided a more representative evaluation of model performance, as the average data presented a smoother and more distinct pattern, as shown in Figure 2b–d. Additionally, the performance metrics provided a quantitative assessment of the average accuracy of each model category. Note that the SMAPE was used in panels a–b while the MPE was used in panel c to better compare the model performance considering the cancellation effect, as discussed in Section 2.5. Moreover, if the best-fit line was above or below the unity line, it indicated that the models overestimated or underestimated the actual PM_{2.5} values, respectively.

Specifically, panel (a) of Figure 3 shows scatter plots comparing the estimated PM_{2.5} concentrations against the true PM_{2.5} concentrations for the averages of the linear, tree, and ANN models, with actual values ranging from 0 to 50 µg/m³. The linear models (top plot) exhibited a moderate correlation with an NNSE of 0.29, an RSR of 1.57, and an SMAPE of 13.75%. The tree models (middle plot) showed improved performance, with an NNSE of 0.73, an RSR of 0.60, and an SMAPE of 11.80%. For the averages of the ANN models (bottom plot), the NNSE improved by 165% compared to linear models and 5% compared to tree models. The RSR improved by 65% compared to linear models and 10% compared to tree models. Additionally, the SMAPE in ANN models improved by 19% compared to linear models and 6% compared to tree models. Overall, the ANN models showed the closest alignment between the best-fit and unity lines, as well as the lowest MAE and SMAPE, indicating that the ANN models' estimates were more accurate than those of the other models. However, the tail of the plot was much longer than the head, making the overestimation less apparent to stakeholders.

Therefore, panel (b) provides scatter plots on a log₁₀ scale for estimated and true PM_{2.5} concentrations ranging from 1 to 100 µg/m³ to visualize the distribution comprehensively. For the linear models, the NNSE was 0.75, the RSR was 0.57, and the SMAPE was 8.24%, indicating better performance on the log scale as it was easier for machine learning models to learn the data pattern. It is also evident that the models tended to overestimate lower concentrations and underestimate higher concentrations, as shown by the best-fit line deviating from the unity line. Tree models performed similarly well on the log scale, with an NNSE of 0.81, an RSR of 0.48, and an SMAPE of 7.12%. The ANN models again demonstrated the best performance, with an NNSE increased by 10.6% and 2.5% compared

to linear and tree models, respectively, an RSR improved by 21.0% and 6.2%, respectively, and an SMAPE improved by 18% and 5.1%, respectively. The best-fit line in the ANN's plot was the most closely aligned with the unity line compared with the other two model categories, indicating less biased estimations across the $PM_{2.5}$ concentration from 1 to $100 \mu\text{g}/\text{m}^3$ on the log10 scale.

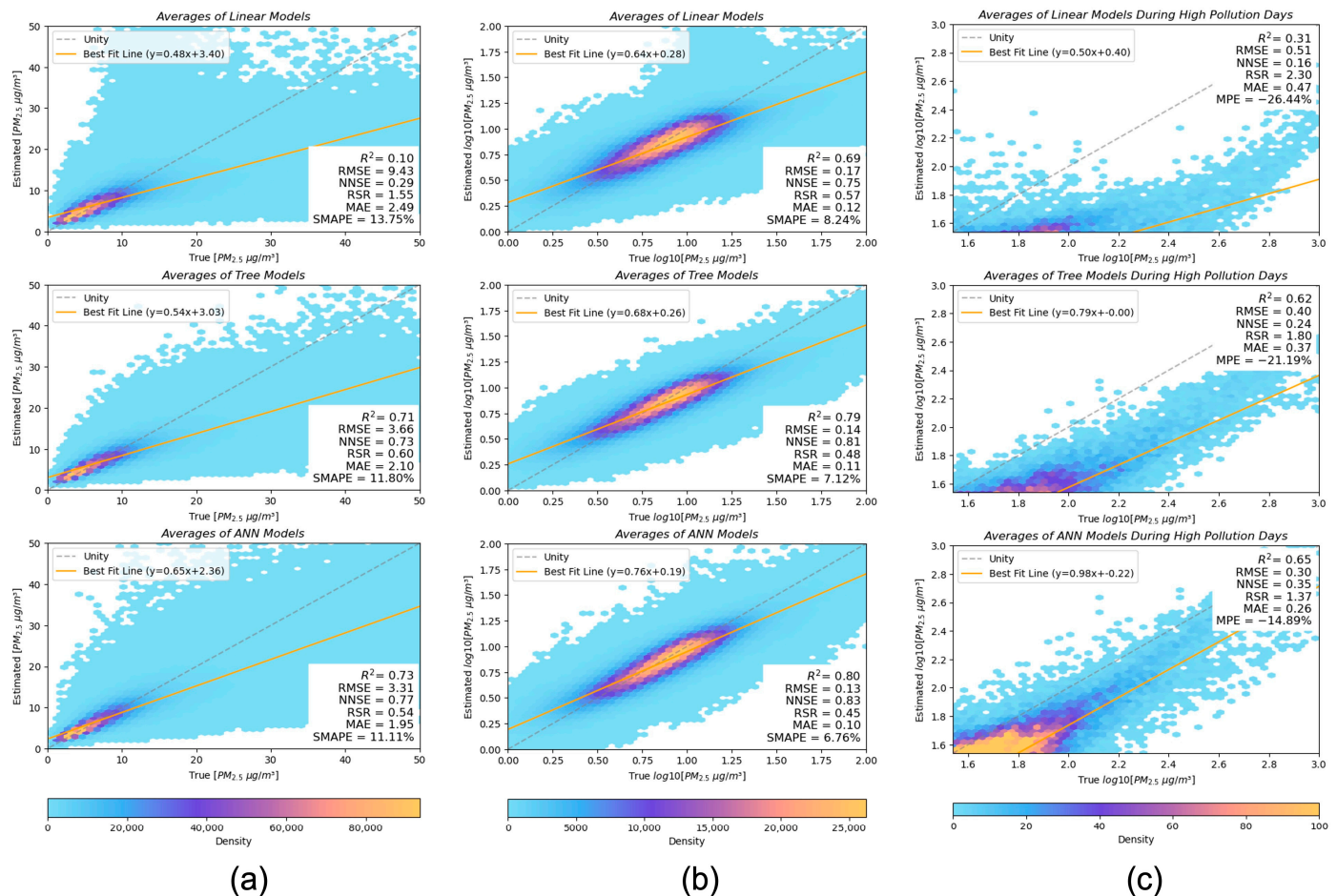


Figure 3. Scatter plots comparing the estimated $PM_{2.5}$ concentrations against the true $PM_{2.5}$ concentrations for averages of linear, tree, and ANN models under 3 scenarios. (a) Estimates from the averages of each model category and the actual values from 0 to $50 \mu\text{g}/\text{m}^3$; (b) estimates from the averages of each model category and the actual values from 1 to $100 \mu\text{g}/\text{m}^3$ on the log10 scale; (c) estimates from the averages of each model category and the actual values during high pollution days from 35 to $1000 \mu\text{g}/\text{m}^3$ on the log10 scale. The color gradient indicates the density of data points. Performance metrics (R^2 , RMSE, NNSE, RSR, MAE, and MPE) are displayed for each model category. The dashed line represents the unity line ($y = x$), and the solid line represents the best-fit line for each model category. The transition between the colors of light blue, medium blue, purple, pinkish purple, orange, and yellow defines the density visual range, moving from low to high density. The shadows in the figure occur in areas of overlapping density where the darker colors (purple, pinkish purple, orange, and yellow) dominate due to the higher concentrations of data.

Panel (c) focuses on high pollution days, with $PM_{2.5}$ concentrations from 35 to $1000 \mu\text{g}/\text{m}^3$ on the log10 scale. The linear models exhibited poor performance under these conditions, with an NNSE of 0.16, an RSR of 2.30, and an MPE of -26.44% . The data points showed significant deviation from the unity line, highlighting the inadequacy of linear models for high pollution levels. Tree models performed better, with an NNSE of 0.24, an RSR of 1.80, and an MPE of -21.19% , but still showed considerable spread. ANN models, while also challenged under high pollution conditions, maintained relatively

impressive performance with an NNSE improved by 119% and 46% compared to linear and tree models, respectively, an RSR improved by 40% and 24%, respectively, and an MPE improved by 44% and 30%, respectively. The points were more closely clustered around the unity line compared to linear and tree models, indicating a higher resilience of ANN models in extreme conditions.

In conclusion, Figure 3 reveals that ANN models consistently outperformed linear and tree models across different scenarios. The NNSE, RSR, and MPE metrics demonstrated that ANN models provided the most accurate and reliable estimates of PM_{2.5} concentrations, particularly in complex and high pollution scenarios. This superior performance can be attributed to the ANN models' ability to capture nonlinear relationships and complex patterns in the data, making them a robust tool for air quality estimation and monitoring. Our findings demonstrate the efficacy of even the simplest neural networks in air quality estimations, especially for high pollution events, challenging the traditional dominance of linear and tree-based models. They also underscore the potential for significant improvements through more sophisticated neural networks, showing promise for further improving air quality data access for public health and environmental policy formulation.

3.5. Spatial Distribution of Annual and Seasonal Mean MPE

To explore the spatial and temporal patterns of model estimations, we plotted the 2-year average and seasonal mean model performance in each grid cell. We selected the MPE as the performance metric here because it effectively illustrated levels of underestimation and overestimation, as demonstrated in previous sections.

In Figure 4, panels a–c correspond to the scenarios depicted in Figure 2b–d, respectively. The first left plot of each panel shows the spatial distribution of actual PM_{2.5} concentrations (true values) from MERRA-2, using different scales to facilitate comparison of the MPE of model-averaged estimations across the contiguous United States. The other figures in each panel display the spatial pattern of annual average MPE for each model category, with red representing overestimation and blue representing underestimation. The ocean is masked in white in each plot.

Panel (a) shows the scenario of filtering the daily true PM_{2.5} values from 0 to 50 µg/m³ without any scales, although the plot shows a range of 0–25 µg/m³ to enhance the color gradient. The MPEs of annual-mean model-averaged PM_{2.5} estimations have positive values in most regions, indicating that machine learning models tend to overestimate PM_{2.5} levels in areas with low and moderate pollution. Comparing the model categories, the red color is lightest in the plot of ANN models and deepest in the plot of linear models, highlighting the advantages of ANN models in estimating PM_{2.5} during low and moderate pollution days. This comparison underscores the consistent overestimation by linear models, which may be attributed to their inability to capture nonlinear relationships in the data, a capability where ANN models excel.

Panel (b) presents the scenario of the most polluted days, filtering the daily true PM_{2.5} values from 1 to 100 µg/m³ with log10 scales. After the log10 transformation, the annual mean model results showed an overall overestimation on the most polluted days, consistent with the analysis in Figures 2c and 3b. Unlike the MPE plots in panels (a), panel (b) shows the error distribution of the averages of the linear models, the MPE differences between linear and tree models, and between tree and ANN models for a better comparison by observing the color gradients. In the Tree Models – Linear Models plot, red regions tend to align with white-blue areas in the linear plot, indicating that tree models generally underestimated PM_{2.5} concentrations less in areas where linear models underestimated.

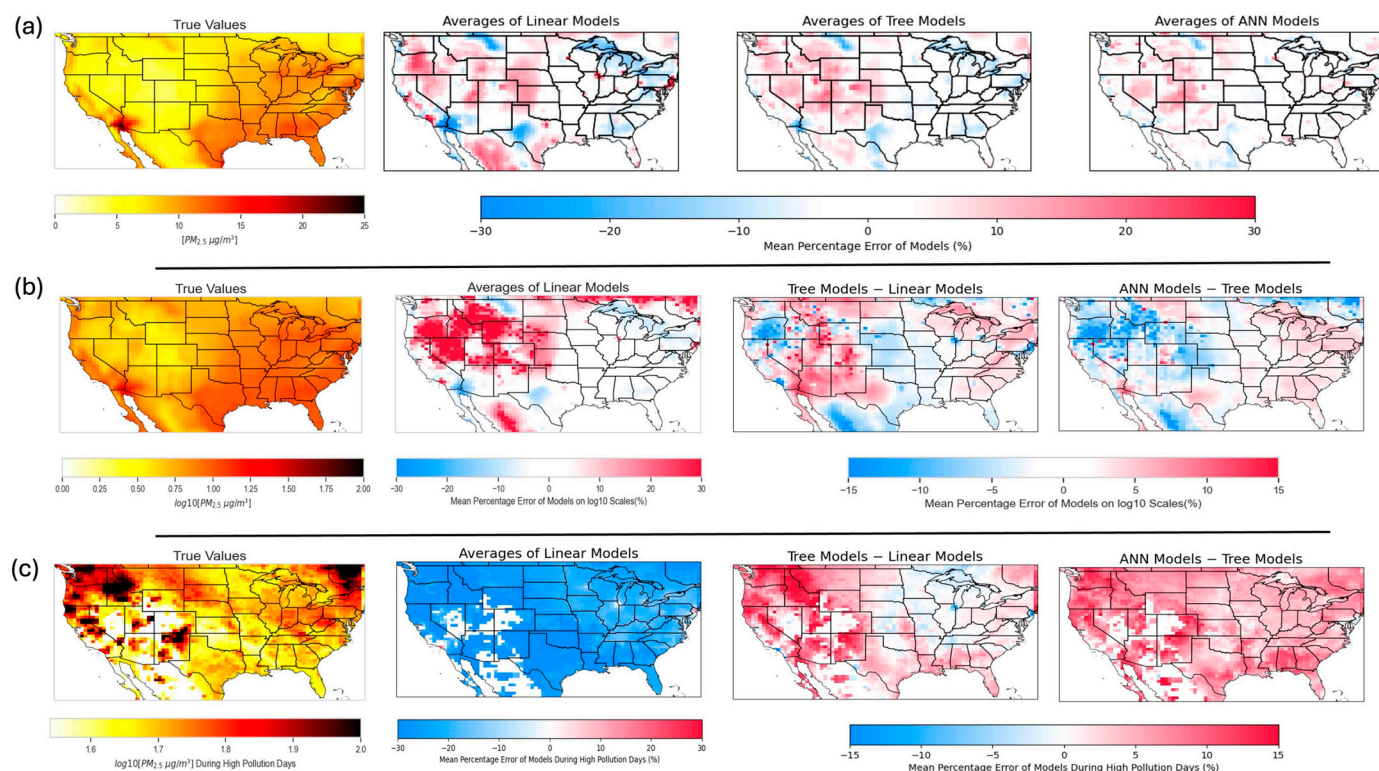


Figure 4. Spatial distribution of 2-year average actual concentrations of $PM_{2.5}$ and the mean percentage error (%) of model-averaged estimated concentrations of $PM_{2.5}$ in the scenarios of (a) actual $PM_{2.5}$ from 0 to $50 \mu\text{g}/\text{m}^3$; (b) $\log_{10} PM_{2.5}$ from 0 to 2 ($1\text{--}100 \mu\text{g}/\text{m}^3$); and (c) $\log_{10} PM_{2.5}$ from 1.54 to 3 ($35\text{--}1000 \mu\text{g}/\text{m}^3$).

Conversely, tree models tended to overestimate less in areas where linear models show red colors. Similarly, in the ANN Models – Tree Models plot, red regions align with white-blue areas in the linear plot, while white-blue areas correspond to red regions in the linear plot. This suggests that ANN models also tended to overestimate less where tree models overestimated less than linear models and underestimated less where tree models underestimated less than linear models.

Overall, both tree and ANN models seemed to adjust for the linear models' tendency to overestimate and underestimate $PM_{2.5}$ concentrations in different regions. This suggests that ANN models maintain their superior performance across different pollution levels, likely due to their capacity to adapt to varying data distributions through complex layer structures.

Panel (c) focuses on high and extremely high pollution days by filtering the daily true $PM_{2.5}$ values from 35 to $1000 \mu\text{g}/\text{m}^3$ with $\log_{10}$ scales. Similar to the MPE plots in (b), panel (c) also shows the error distribution of the averages of the linear models, the MPE difference between linear and tree models, and between tree and ANN models. The spatial MPE distribution of the averaged linear models presented an overall underestimation of $PM_{2.5}$. While the results in the averaged tree and ANN models also showed an underestimation, as discussed in Figure 2d, the color gradient between model categories is not distinct. To compare the annual mean MPE between the averaged linear and tree models, a difference plot is provided. This plot reveals that most regions are red, indicating that the averaged tree models underestimated less than the linear models. Similarly, the difference plot between the averaged tree and ANN models shows that most regions are red, indicating that the averaged ANN models underestimated less than the tree models.

Overall, panel (c) demonstrates that ANN models performed the best in estimating $PM_{2.5}$ during high pollution days. The underestimation observed in all models during high pollution days might be due to their limitations in capturing extreme events, yet the ANN models' relatively better performance highlights their robustness in handling severe pollution levels compared to linear and tree models.

In terms of the seasonal-mean analysis, Figure 5 shows the seasonal average MPE for linear models, the difference between tree and linear models, as well as the difference between ANN and tree models across the contiguous United States for spring, summer, fall, and winter during high pollution days ($PM_{2.5}$ from 35 to 1000 $\mu g/m^3$ with log10 transformation). As discussed in Figure 2d, all models underestimated the actual $PM_{2.5}$ concentrations. While the blue color in the first plot from panels a to d shows that linear models significantly underestimated $PM_{2.5}$ concentrations in most regions with high pollution during all seasons, the red (or blue) color in the difference plots (Tree Models – Linear Models or ANN Models – Tree Models) indicates less (or more) underestimation. The white color in the plots means that there were no high pollution events or differences between models.

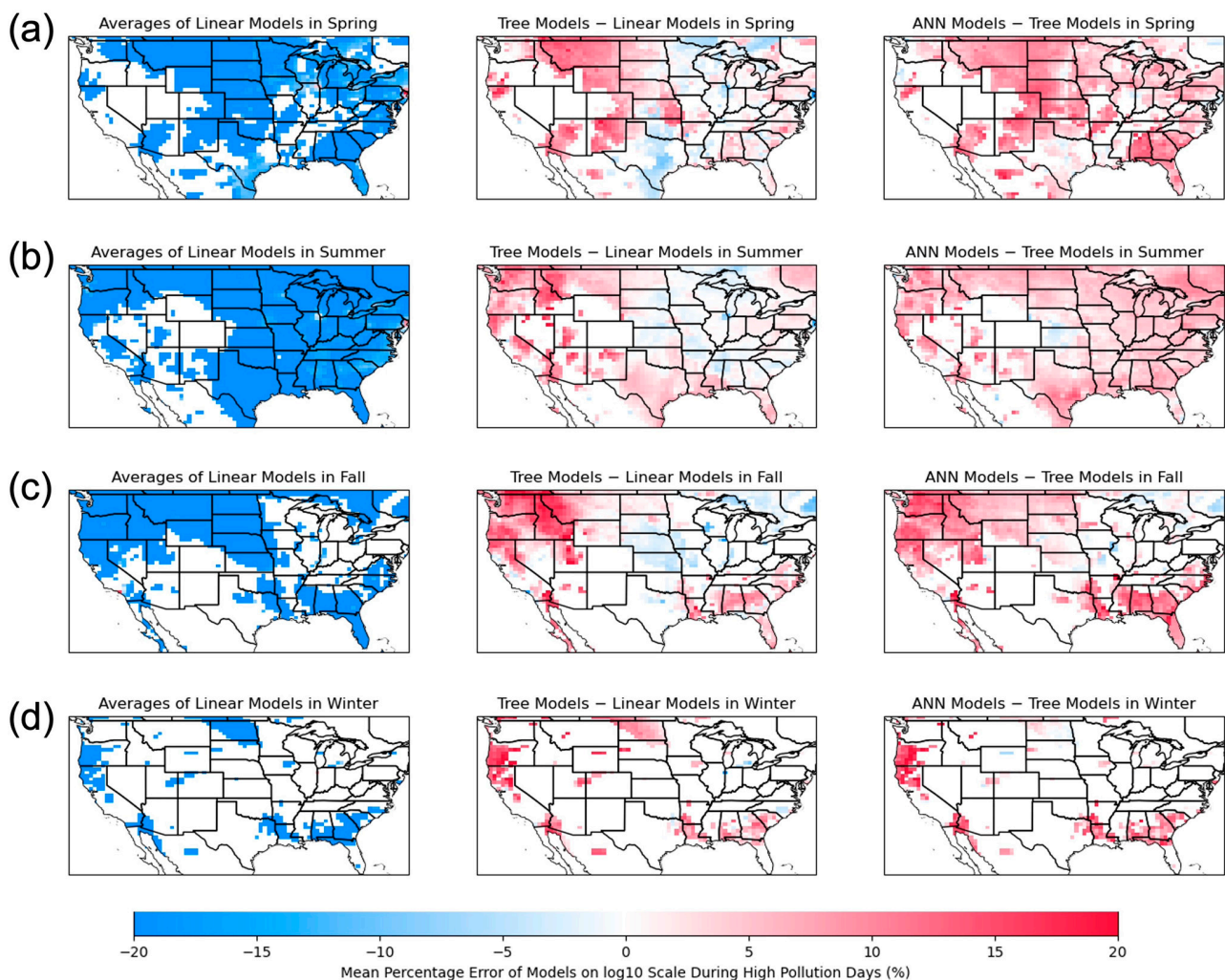


Figure 5. Spatial distribution of the seasonal mean percentage error (%) of model-averaged estimated concentrations of $PM_{2.5}$ in the scenario of $PM_{2.5}$ from 35 to 1000 $\mu g/m^3$ with log10 transformation. Panels (a–d) represent spring, summer, fall, and winter, respectively. The first plot in each panel represents the performance of linear models at each location; the middle plot shows the error difference between tree and linear models; and the last plot shows the error difference between ANN and tree models.

Therefore, in spring (panel a), the Tree Models – Linear Models plot, primarily red, suggests that tree models generally reduced this underestimation across most areas, although some states like Texas, Minnesota, Iowa, Wisconsin, and Illinois still show light blue, indicating their weaker performance. The ANN Models – Tree Models plot, however, is almost entirely red, indicating that ANN models further improved accuracy, reducing underestimations even more effectively than tree models. While the difference between ANN and linear models is less apparent in the blue areas of the Tree Models – Linear Models plot, this pattern shows that ANN models better captured $PM_{2.5}$ variations overall in spring.

In summer (panel b), the Tree Models – Linear Models plot with most red areas in the western U.S., Texas, and parts of the eastern U.S. indicate the tree models underestimated less in these regions, while some regions in Kansas, Missouri, and Tennessee are in light blue colors indicating the worse performance of tree models. Nonetheless, all regions in the ANN Models – Tree Models plot are red except the state of Kansas, which proves the superiority of using ANN models to estimate summer $PM_{2.5}$.

In fall (panel c), the Tree Models – Linear Models plot followed a similar red pattern to panel b but with blues in the Kansas, Nebraska, Iowa, and Missouri regions. Moreover, ANN models obtained an overall outstanding performance in estimating $PM_{2.5}$ in fall from the observations of the ANN Models – Tree Models plot.

In winter (panel d), tree models underestimated $PM_{2.5}$ concentrations less in most regions from the Tree Models – Linear Models plot but still performed worse compared with ANN models from the ANN Models – Tree Models plot.

Overall, the seasonal analysis revealed that ANN models consistently outperformed linear and tree models across all seasons. The linear models tended to show significant underestimations, especially in the Western and Southeastern U.S., likely due to their inability to capture nonlinear relationships in the data when climate change had more impacts there. Tree models, while performing better than linear models, still exhibited considerable underestimations. ANN models demonstrated the most balanced performance, effectively capturing the seasonal variations in $PM_{2.5}$ concentrations and showing fewer areas of significant error. This comprehensive analysis highlights the robustness and adaptability of ANN models in different seasonal contexts on the background of climate change.

3.6. Global and Local Sensitivity Analysis of Features in Tree and ANN Models

Following the comprehensive performance evaluation of ML models, this sub-section validates these models through a sensitivity analysis to reconfirm the feature selections from global and local perspectives. A global sensitivity analysis helps determine how different values of input features impact the overall model's estimations, thereby identifying which features are most influential. This validation is crucial for solidifying the foundations of our models and enhancing their interpretability, especially for complex models like ANN and tree models. In tree models, feature importance analysis, which measures the contribution of each feature to the model's estimations, is a form of sensitivity analysis. Similarly, in ANN models, permutation methods, which assess the impact of changes in input features on the output, serve as another form of sensitivity analysis. By understanding the global sensitivity of each feature, we can ensure that the selected features contribute significantly to the model's overall reliability and interpretability. On the other hand, a local sensitivity analysis explores the relative impact of features on individual estimation. SHAP values are a popular method for this type of analysis because they are calculated based on cooperative game theory and provide a way to fairly distribute the contribution of each feature to the estimation from a local perspective [75]. In this section, the spatial distributions of SHAP values are only used to explain how much each feature contributes to the difference

between the actual estimation and the average estimation at each location. The positive or negative SHAP values indicate the feature can increase or decrease the average estimation at a certain location.

Specifically, Tables 6 and 7 were generated based on the inherent ability of tree and ANN machine learning algorithms to determine the importance of each feature in the training or test set. The feature names in Tables 6 and 7, along with their full names and descriptions, are listed in Table 1. The sensitivity analysis of features in tree models is a result of the algorithm's fitting routine, often as a byproduct of the decision-making process within the tree structure, typically derived from each feature's contribution to impurity reduction, where the impurity here refers to a measure of the disorder or randomness in a dataset at a given node in tree-based models. For the sensitivity analysis of features in ANNs, the permutation feature importance method was employed. This method involves measuring the change in the model's performance when the values of a feature are randomly shuffled [30]. By permuting the feature values, its relationship with PM_{2.5} is broken, and the impact on the model's performance indicates the importance of that feature. The larger the decrease in R², the more important the feature is considered to be. This approach provides a way to assess feature importance in ANNs, which are otherwise challenging to interpret due to their black-box nature [30]. By comparing the model's performance before and after permutation, we can infer which features are critical for making accurate estimations.

Table 6. Rankings of feature importance scores across tree-based models, which are inherent to the model's fitting process. The rankings exclude the AdaBoost model due to its considerably underperforming others. The specific descriptions of each feature are listed in Table 1.

Feature Names	Long Name of Features	Random Forest	Extra Trees	Gradient Boost	XGBoost	Average Rank
TOTEXTTAU	Total aerosol extinction AOD at 550 nm	1	1	1	1	1
TOTABSTAU	Total aerosol absorption AOD at 550 nm	2	2	3	2	2
COSC	Surface CO concentration	3	6	2	4	3
T2M	2 m air temperature	5	4	5	3	4
QV2M	2 m specific humidity	4	3	7	5	5
lat	Latitude	7	5	4	6	6
PS	Surface pressure	6	7	8	7	7
lon	Longitude	10	8	6	9	8
O3	Surface O ₃ concentration	9	10	10	8	9
V2M	2 m northward wind	8	11	9	10	10
MonthCos	Cosine component of the month	13	9	12	12	11
ALBEDO	Surface albedo	11	13	14	13	12
SLP	Sea level pressure	12	12	13	14	12
DISPH	Zero-plane displacement height	15	15	11	11	14
EMIS	Surface emissivity	14	14	15	15	15

Table 7. Rankings of permutation feature importance across ANN models, which was measured by evaluating prediction changes when features were shuffled. The specific descriptions of each feature are listed in Table 1.

Feature Names	Long Name of Features	ANN1	ANN2	ANN3	ANN4	Average Rank
TOTEXTTAU	Total aerosol extinction AOD at 550 nm	1	1	1	1	1
QV2M	2 m specific humidity	2	2	3	2	2
TOTABSTAU	Total aerosol absorption AOD at 550 nm	3	3	2	3	3
T2M	2 m air temperature	4	4	4	4	4
COSC	Surface CO concentration	5	5	5	5	5
PS	Surface pressure	7	6	6	6	6
lat	Latitude	6	7	7	7	7
lon	Longitude	8	8	8	8	8
MonthCos	Cosine component of the month	9	9	9	9	9
V2M	2 m northward wind	10	10	10	10	10
SLP	Sea level pressure	12	11	11	11	11
O3	Surface O ₃ concentration	11	12	12	12	12
ALBEDO	Surface albedo	13	13	13	14	13
EMIS	Surface emissivity	14	15	14	13	14
DISPH	Zero-plane displacement height	15	14	15	15	15

In contrast to the feature selection procedure described in Sections 2.2 and 3.1, the rankings herein reflect the actual importance of each feature post-training. These importance scores indicate the degree to which each feature enhances the model's estimation accuracy. It is crucial to note that the feature importance extracted here is not directly comparable across different models, as they are calculated based on criteria specific to each model's algorithm. To enable a standardized comparison, the feature importance from each model was converted into rankings. Then, the ranks attributed to each feature by models were averaged, yielding a single ranking per feature, quantifying the feature's relative significance across all the tree-based and ANN models. This sensitivity analysis revalidating feature selection reinforced model validation and deepened our understanding of the estimation system, enabling further model refinement and practical application in the future.

Specifically, Table 6 shows a comparative view of feature importance across four tree-based models used in this study. The total aerosol extinction AOD at 550 nm stood out as the most influential feature across all models, consistently achieving the top rank. The features of total aerosol absorption AOD at 550 nm and surface CO concentration also commanded significant importance with average ranks of two and three, respectively. Their high ranking across various models signified the importance of AOD and trace gas features. On the contrary, features like zero-plane displacement height and surface emissivity were ranked lowest in terms of feature importance at 14th and 15th, respectively, indicating less significance of topography-relevant features. Moreover, features such as latitude/longitude, surface pressure, and surface O₃ concentration moderately impacted PM_{2.5} estimations, with their average ranks ranging from 6th to 9th. This indicated the importance of spatial variance and other air pollution.

Additionally, Figure 6 shows the specific feature importance scores across tree-based models. Each bar represents the importance score of a particular feature, calculated by the model's inherent feature importance method. The pale (lightest) green indicates the XGBoost model, the light green represents the gradient boosting model, the forest green represents the extra trees model, and the deep green represents the random forest model. The features are arranged in order of their cumulative importance across all models, with the most significant features (total extinction AOD) positioned at the top. The ranking result closely aligned with the qualitative analysis ranking presented in Table 6, with the exception that the surface albedo was ranked 14th here. Furthermore, the lengths of the bars in the gradient boosting model were significantly different from those of the XGBoost, extra trees, and random forest models, which exhibited relatively similar distributions.

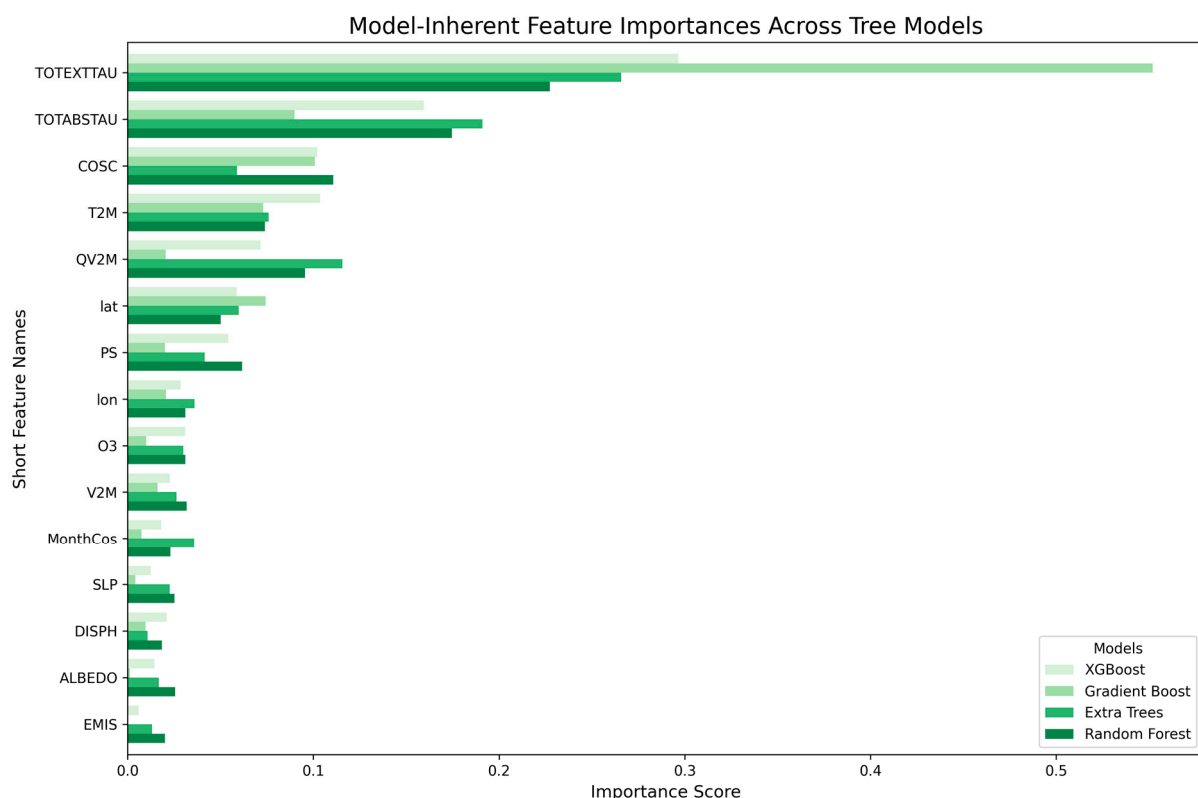


Figure 6. Feature importance scores across four tree-based models (random forest, extra trees, gradient boosting, and XGBoost). Each bar represents the importance score of a specific feature as calculated by the model's inherent feature importance method. The scores represent the relative contribution of each feature to the models' predictions. Features are sorted in descending order of their total importance across all models, with grouped bars showing the individual contributions from each model for direct comparison.

In short, Table 6 and Figure 6 show that some features were universally recognized as critical estimators, and some others were considered less impactful across different tree-based modeling approaches. It sets the stage for further inquiries into why certain features are more crucial and how various models could be integrated based on these insights to refine model performance. The feature importance derived from the tree models post-training generally affirmed that the mutual information and Spearman correlation successfully guided feature selection. Nevertheless, the ranking based on Spearman correlation and MI (before the training) could not entirely reflect the full importance of some features, such as surface O₃ concentration and longitude, that were ranked much higher in the post-training feature importance ranking. This is because the tree-based model

importance is model-specific and can vary depending on the structure of the tree and the hyperparameters used, accounting for interactions and the model's structure. In contrast, MI and Spearman correlation methods are model-agnostic measures and do not depend on any specific estimation model.

Table 7 shows the results of assessing feature importance in ANN models using the permutation feature importance method [30]. The process started by calculating R^2 on the unshuffled data. Then, one feature at a time was shuffled across all samples in the training dataset, and the model's R^2 was recalculated. The importance of the feature was determined by the difference between performance in the baseline and the performance with the shuffled feature; a larger decrease (degradation in performance) indicated a more important feature. The shuffling process was randomly repeated 30 times for each feature to average out variability in the results.

Consistently, Table 7 indicates that total aerosol extinction AOD at 550 nm ranked as the most important feature across all ANN models, followed by 2 m specific humidity, total aerosol absorption AOD at 550 nm, and 2 m air temperature. The feature zero-plane displacement height and surface emissivity were ranked as the least important features across all models, which was the same as their contribution to tree models. Compared with the feature importance rank across tree-based models, there was some consistency, with extinction AOD being highly important, but other features showed variations in their rankings. For instance, the temporal feature was ranked lower in tree-based models than in ANNs, which indicates that the feature Month interacted with the $PM_{2.5}$ concentration in ways that ANN models captured more effectively, while ANN models could also capture the spatial interactions with $PM_{2.5}$ effectively. The better performance of ANNs thus further suggests that future work needs to incorporate data with temporal/spatial information or to develop more complex models with such information as convolutional neural networks that better capture the spatial variance of $PM_{2.5}$. Moreover, tree-based models often handle nonlinear relationships differently than ANNs, which could explain why some features vary in importance across model types. Such differences underline the importance of model selection in machine learning, as different models may interpret the importance of features differently based on their underlying algorithms.

Additionally, when we compared the results in Table 7 to the initial feature importance ranking in Table 3, we found that MI scores had extinction AOD as the most important feature, which was consistent with the ANN's ranking after training. However, MI scores listed features like 2 m northward wind and surface O_3 concentration as less important than the ANN's post-training ranking suggested. Conversely, features such as surface emissivity and albedo, which were highly rated by MI scores, were ranked lower after training. This discrepancy can arise because the MI score measures theoretical dependency between features and the target variable, while the ANN post-training rankings reflect the practical utility of the features within the context of a particular model's architecture and learned patterns. It shows that pre-training importance metrics might not always correspond to how a model will weigh features after training, emphasizing the complex nature of feature interactions within neural networks.

In terms of the local sensitivity analysis, Figure 7 shows the spatial distribution of SHAP values for the overall top six important features in Tables 6 and 7 but derived from the XGBoost model. We used this model instead of extra trees or ANN models to conduct the local sensitivity analysis because XGBoost has higher computational efficiency and has been widely used in recent studies, e.g., [76–78].

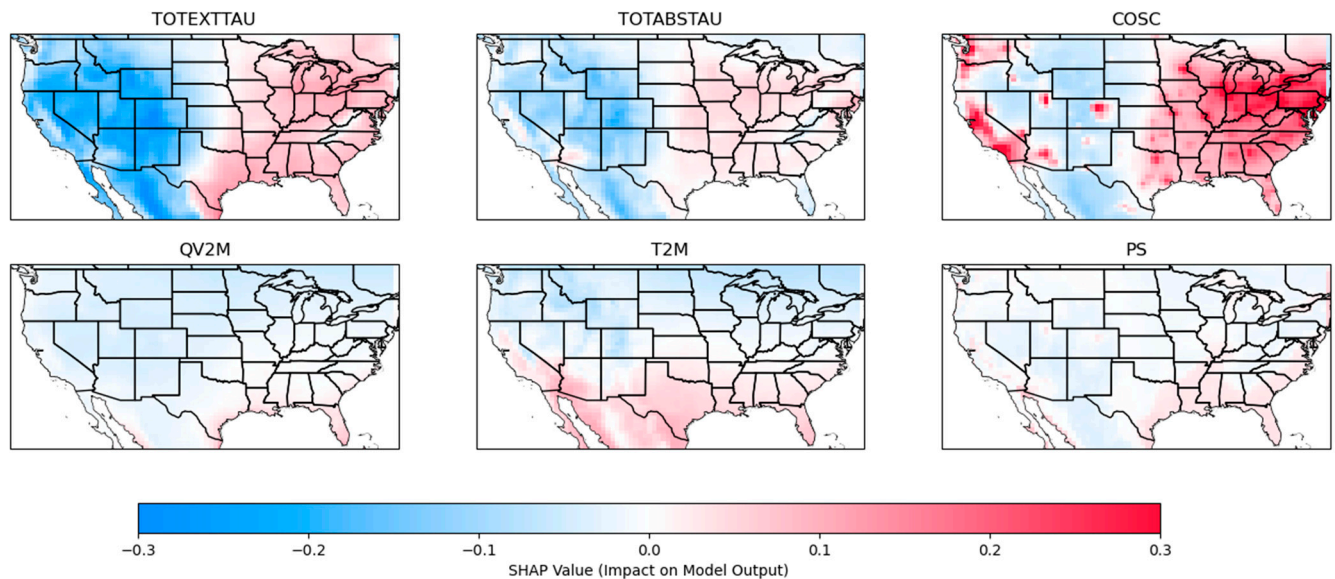


Figure 7. Spatial distribution of the 2-year average SHAP values at each location under the scenario when $\text{PM}_{2.5}$ ranges from 1 to $100 \mu\text{g}/\text{m}^3$. The positive SHAP values (red areas) indicate that the feature can increase the average estimated $\text{PM}_{2.5}$ concentrations, while the negative SHAP values (blue areas) indicate that the feature can decrease the average $\text{PM}_{2.5}$ estimations at certain locations. The white areas mean the feature has non-distinct impacts on the estimations.

While the SHAP values in Figure 7 show a clear positive or negative pattern at each location, it is important to note that models overestimated the actual $\text{PM}_{2.5}$ concentrations in the Western United States and underestimated it in the eastern regions, as discussed in Figure 4. Thus, the blue color in Figure 7 indicates that the feature can help reduce the overestimation if the model in that location overestimates the pollution. Similarly, the positive values (red areas) indicate that the feature can help reduce underestimation if there are underestimations.

Taking the SHAP distribution of the total extinction AOD at 550 nm (TOTEXTTAU) as a specific example, negative SHAP values are shown in the Western U.S., where the model generally overestimated $\text{PM}_{2.5}$ concentrations. We can conclude that the total extinction AOD has negative impacts on the model output or the value of $\text{PM}_{2.5}$ estimation, but we cannot conclude that this feature has negative impacts on model performance. Nonetheless, it helps improve the performance by reducing the overestimations.

Therefore, the features of total extinction and absorption AOD show similar balanced importance patterns, and the total extinction AOD is more important than the total absorption AOD at all locations by observing the color gradient. Furthermore, while the SHAP values show that surface CO concentrations are less important in the western region, they emphasize the highest importance of CO in reducing underestimation in the eastern region compared to other features, which highlights the importance of using trace gas features to estimate surface $\text{PM}_{2.5}$ concentrations. Also, the difference in the SHAP's spatial pattern of 2 m specific humidity and temperature reconfirms our decision not to filter either of them during the feature selection process. Although features in the second row in Figure 7 show different patterns from features in the first row, they still make certain contributions to estimate surface $\text{PM}_{2.5}$ on most days.

Overall, our global and local sensitivity analysis of features in tree and ANN models enhanced the interpretability of our models and increased their robustness in estimating $\text{PM}_{2.5}$ concentrations over the continental United States. The analysis reconfirmed the importance of using total extinction and absorption AOD at 550 nm, surface concentrations of CO, surface pressure, and 2 m specific humidity and temperature to decrease the overes-

timation and underestimation, especially for the importance of surface CO concentrations in the Eastern United States.

4. Discussion

The findings from this comprehensive evaluation of machine learning models highlighted significant insights into estimating surface PM_{2.5} concentrations, especially for extreme pollution events when daily PM_{2.5} is equal to or larger than 35 µg/m³. The study demonstrated the relative strengths and limitations of various statistical and ML models, emphasizing the notable performance of ANN models, the simplest NN model. This discussion delves into the results' implications, situates them within the broader context of existing research, and outlines potential future research directions.

The analysis in the result section revealed that ANNs consistently outperformed traditional linear models and tree-based models across different pollution levels and seasons. This superior performance of ANNs can be attributed to their ability to capture complex, nonlinear relationships between PM_{2.5} and its influencing factors based on the sensitivity analysis of features. The higher R², lower RMSE, and MAE values, especially in ANN3 architectures, underscore the efficacy of ANNs in handling high-dimensional and noisy data. When dealing with long-tailed data distributions, such as those found in PM_{2.5} concentration datasets, it is crucial to use performance metrics that can adequately capture the nuances in the data. The study employed several normalized metrics, including the NNSE, the RSR, the SMAPE, and the MPE, which provide robust measures of model performance in the presence of skewed data at different annual levels.

However, one may be concerned that the R² scores of our models did not exceed 0.9. This is because the scope of our study considered all extreme pollutant events (outliers) and covered a large spatial domain (national) using one global model, as illustrated in Figures 2–5. These factors inherently introduced greater complexity into our models, negatively impacting the overall performance metrics. In contrast, some studies reporting higher R² values typically focus on a smaller regional domain (e.g., [32–34,79]) with less emphasis on the higher pollution events that are important for public health (e.g., [32–34]).

While the differences in R² between the ANN and other models were also small when averaged across the entire national domain over 2 years, it was evident in Figure 2b–d that the ANN curve aligned more closely with the actual PM_{2.5} curve during both low and high pollution days, particularly at the right tails of the curves. In Figure 3c, ANN models improved the R² by 110% compared with linear models and 5% compared with tree models. In Figures 4 and 5, the distinct red color gradient from linear models to ANN models indicated that ANN models performed the best in estimating PM_{2.5} on most days and high pollution events across all seasons. Nonetheless, we agree that the ANN models should be further improved in future studies by integrating data with more information and advancing the ANN model to more complex deep learning approaches that account for spatial and temporal variations to achieve distinctly higher R².

In addition, while most current studies (e.g., [32–34]) do not specifically discuss the model performance in estimating PM_{2.5} during high pollution events when daily PM_{2.5} ranges from 35 to 1000 µg/m³, our results revealed that there were much higher PM_{2.5} concentrations in the Western U.S., with less high concentrations in the southern and eastern regions. The annual and seasonal patterns of averaged model estimations suggested that linear models exhibited significant underestimations in most regions and seasons. Tree models also showed widespread underestimations, but with less intensity, except for some states (e.g., Kansas) in the summer and fall. ANN models displayed a more consistent improvement in MPE, indicating better performance across different regions and seasons compared with linear models and tree models.

Moreover, while Sayeed et al. [79] concentrated on high pollution events, their methodology incorporated chemical components of PM_{2.5} from MERRA-2 for estimations, which undermines the objective of estimating PM_{2.5}, as these values facilitate the direct computation of total PM_{2.5}. Furthermore, their study was confined to PM_{2.5} estimation at a regional scale. Our systematic approach, which excluded the use of these chemical components and encompassed a national scale, addressed these discrepancies by ensuring consistent data quality and coverage. Consequently, this methodology offers a more robust comparison of model performances and estimations of PM_{2.5} concentrations during high pollution days.

Therefore, the encouraging performance of ANNs in this study, especially under the high pollution scenario, suggests a pathway toward more accurate and reliable air quality estimates, which are crucial for long-term exposure analysis and public health studies. The ability of ANNs to handle complex data structures and capture subtle patterns in PM_{2.5} concentrations can significantly enhance the precision of exposure assessments and epidemiological research.

Future research should explore more sophisticated neural network architectures, such as a two-dimensional convolutional neural network model with spatial information and a recurrent neural network (e.g., long short-term memory) with temporal information. This can further improve the model performance as well as the spatial and temporal resolution of PM_{2.5} estimations. Integrating multi-source data, including chemical transport models, satellite observations, and ground measurements, could also help enhance model robustness and accuracy.

Additionally, combining global and local sensitivity analyses using three different methods significantly improved our model's robustness and helped confirm the most important features when making estimations. It is impressive that the trace gas feature of surface CO played an important role in estimating the eastern surface PM_{2.5} concentrations. However, the sensitivity analysis of features, or the interpretability of ML models, remains a critical area of development. The computation of SHAP values for deep learning models with extremely large datasets still needs to be explored in future work to provide deeper insights into model behavior and feature contributions at each location, thereby improving the transparency and applicability of these models in policy-making and environmental management.

Overall, this study underscores the significant potential of ANN models in estimating surface PM_{2.5} concentrations, focusing on the high pollution events in the continental United States. By providing a comprehensive comparison and highlighting the importance of feature selection and the sensitivity analysis of features, it sets the stage for future advancements in estimating air pollutants, including PM_{2.5}, and other pollutants like PM₁₀, SO₂, and NO₂.

5. Conclusions

In conclusion, this study conducted a comprehensive evaluation of various machine learning models for estimating surface PM_{2.5} concentrations, focusing on high pollution events in the continental United States using the MERRA-2 dataset. The MERRA-2 dataset provided comprehensive pollutant and meteorological data for our study area, offering significant advantages over ground-based observations, which can be limited in spatial and temporal coverage. This allowed for more extensive and continuous monitoring on a national scale. The results revealed the superior average performance of ANNs compared to traditional linear models and tree-based models, especially on high pollution days when daily PM_{2.5} was larger than 35 µg/m³, with an R² improved by 110% and 5% compared to linear and tree models, respectively, an RMSE improved by 41% and 25%, respectively, an MAE improved by 45% and 30%, respectively, an NNSE improved by 119% and 46%,

respectively, an RSR improved by 40% and 24%, respectively, and an MPE improved by 44% and 30%, respectively. When comparing the performance of individual models, ANN3, with an NNSE of 0.84, an RSR of 0.55, and an SMAPE of 11.29%, also outperformed other models. The findings from the spatial distribution of MPE indicated that ANNs could effectively capture the complex relationships between extremely high PM_{2.5} and its influencing factors, leading to more accurate and reliable air quality estimations under high pollution scenarios across seasons.

The comparison with previous studies reinforced the potential of ANNs in air quality estimation on a national scale. This study highlighted the importance of consistent data quality and coverage in evaluating the model's performance. Moreover, this study uniquely focused on high pollution events, which are underrepresented in other studies that primarily consider more frequent days with moderate pollution levels. High pollution events have substantial implications for public health, making the findings particularly relevant. Additionally, trace gas features like surface CO concentrations showed their relative importance in certain regions from our sensitivity analysis of features and the spatial distribution of SHAP values.

However, there are limitations in this study, including the less-than-ideal model performance measured by the overall R², which reflects the challenges of estimating air quality during high pollution events over a large spatial domain. Nonetheless, given the scope and complexity of this study, these metrics are satisfactory and comparable with other recent studies (e.g., [79]). It is believed that there is significant potential for future improvement by advancing the models, using the sequence of data with spatial and temporal information as input, or developing more complex deep-learning models that incorporate such information, and by utilizing longer records from recent satellite missions over the last decade.

Overall, this study provides valuable insights into the strengths and limitations of different ML models and underscores the importance of feature selection, sensitivity analysis of features, and evaluation in improving the accuracy of PM_{2.5} estimations. The findings have significant implications for public health research and the development of more sophisticated air quality monitoring systems focusing on other pollutants such as PM₁₀, SO₂, and NO₂ during high-pollution days, along with utilizing a longer duration of the dataset or applying different ML models in different regions.

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